UNIT I

1. DC CIRCUITS

1.1 BASIC DEFINITIONS

1.1.1 Charge (Q or q):

Charge is the characteristic property of the elementary particles of the matter. The elementary particles are electron, proton and neutron.

There are two types of charges in nature. They one positive and negative charges.

The symbol of charge is ‘Q’ or ‘q’.

The unit of charge is **Coulomb**.

1 coulomb = charge on 6.28 X 10^{18} electrons.

1.1.2 Electric Current (I):

The flow of free electrons in a metal is called **Electric Current**. The unit of Current is “Ampere” A.

Or

The flow of free electrons in any conductor is called **Electric Current**.

It is also defined as the rate of change of charge with respect to time.

$$\text{(ie) Current (I)} = \frac{Q}{t}.$$

or

$$I = \frac{dq}{dt}.$$

Q is measured in Coulomb.

t is measured in Seconds.

1.1.3 Voltage (V):

The difference in potential energy between the charges is called potential difference. In electrical terminology the potential difference is called voltage.

or

The potential difference between two points in an electric circuit is called voltage.

The unit of voltage is volt. It is represented by “V” or “v”.

1.1.4 Electro-motive force (emf):

It is the force which causes to flow the electrons in any closed circuit.

The unit of electro motive force is Volt.

1.1.5 Power:

Power is the rate of doing work. In other words, power is the work done per unit time.

$$\text{Power} = \frac{\text{work done}}{\text{time}}$$

The unit of power is joules/sec or **watts**.
Power can also be expressed as (kw) **kilo watts**.
Some times power is measured in **Horse power (HP) = 735 watts**. **(Mechanical Horse power)**

$$\text{Power } P = VI \text{ watts (or) } P = E.I$$

$$\text{(or) } P = (IR)I = I^2R \text{ watts. } \because V = IR$$

$$\text{Also, } I = \frac{V}{R}$$

$\therefore$ power, $P = V\left(\frac{V}{R}\right) = \frac{V^2}{R}$ watts.

Note:-

1) 1 KW = 1000 watts.
2) 1 HP = 746 watts. **(Electrical Horse power)**

1.1.6 Energy:

Energy is the capacity to do work.

$$\text{Energy} = \text{Power} \times \text{time} = p \times t$$

Unit of electric energy is **watt sec.** or **joules**.
Since the watt-sec. is too small for practical use, the Kilo Watt Hour (kwh) is generally used.

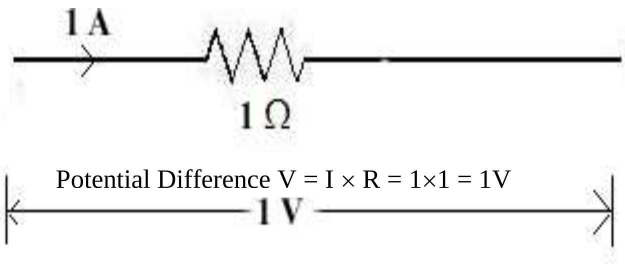
1.1.7 Torque (T):

Torque is the twisting moment or turning moment of a circulating body.
The Unit of torque is Newton-meter. [1kg-m = 9.81 N-m].

1.1.8 Resistance (R):

The opposition offered by a substance to the flow of electric current is called Resistance.

- Resistance may also be defined as the physical property of a substance due to which it opposes (or) restricts the flow of electricity through it. The Unit of resistance is **Ohm** (Ω).
- A device has a resistance of 1 Ohm if a current 1 A flows when a potential difference of 1 V exists across it.



1.1.9 Conductance (G):

Conductance is the inducement to the flow of current, while resistance is the opposition to the flow of current. Hence, it is the reciprocal of resistance.

i.e., $(G = 1/R)$

The Unit of conductance is mho ℧.

1.2 OHM'S LAW:

This law states that "the current flowing through a conductor is directly proportional to voltage across the conductor, provided the temperature remains constant."

"At constant temperature, the potential difference between the two ends of a conductor is directly proportional to the current flowing through the conductor."

$$ie.,\ I \ \alpha\ V$$

$$\boxed{V = IR}$$

Where, R is the resistance of the conductor.

The relationship between potential difference (V), Current (I) and resistance (R) in a d.c circuit is first discovered by the scientist "**George Simon Ohm**" is called ohm's law.

1.2.1 Limitation of Ohm's law:

i. This law is not applicable for all non-metallic conductors or semi conductors.
ii. Not applicable for non-linear device such as zener diodes, arc lamps etc.
iii. If the temperature changes, ohm's law is not applicable for metallic conductors also.

1.2.2 Special note on Ohm's Law:

Ohm's law can be expressed in three forms:

$$i.e., \ \frac{V}{I} = R;$$
$$V = IR;$$
$$I = \frac{V}{R}.$$

1.3 Classification of Circuit Elements:

1. Active elements and 2. Passive elements.

1) *Active elements:*

These elements supply Voltage or Current to the circuit to operate it. They are Voltage sources and Current sources.

Examples: Generators, Transistor, Vaccum tubes etc.,

2) *Passive elements:*

These elements either dissipate energy in the form of heat or store energy from the external sources.

Examples: Resistor (R), Inductance (L) and Capacitor (C).

Resistor (R)	-	Dissipate energy in the form of heat.
Inductor (L)	-	Stores energy in its magnetic field.
Capacitor (C)	-	Stores energy in its electrostatic field.

Problems:

1.1 When a current flowing to a conductor is 2 Amps and the resistance is 5Ω. Find the voltage across the conductor?

Given:

$R = 5\Omega$

$I = 2A$

To find V =?

Formula, ***V = IR*** [As per ohm's Law]

$$V = 2 \times 5$$
$$V = 10V$$

1.2 Find the resistance value when the current flows at 3 amps and 18 volt?

Given:

$V = 18\ volt$

$I = 3\ amps.$

To find R =?

As per ohm's law,

$$R = \frac{V}{I} = \frac{18}{3} = 6\ \Omega.$$

$$\therefore\ R = 6\ \Omega.$$

1.3 Find the current flowing through a conductor. Resistance is 20 Ω and Voltage is 40 V.

Given:

R = 20Ω, V = 40 V, I = ?

$$I = \frac{V}{R} = \frac{40}{20} = 2A.$$

1.4 A resistor carrying a current of 5A has a voltage of 6 V across it. What is its resistance and what power does it dissipate?

Given:

V = 6 V

I = 5 A

Solution:

Resistance, $R = \frac{V}{I} = \frac{6}{5} = 1.2\Omega.$

$P = VI$

Power = 6 × 5 = 30w

$P = 30w$

Also,

$$P = I^2 R$$

$= 5^2 \times 1.2$

$= 25 \times 1.2$

= 30 watts.

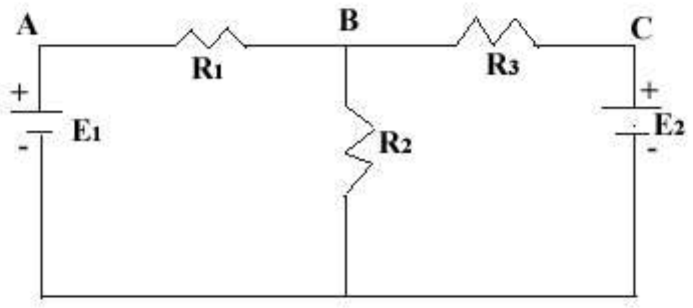
(or)

$$P = \frac{V^2}{R} = \frac{6^2}{1.2}$$

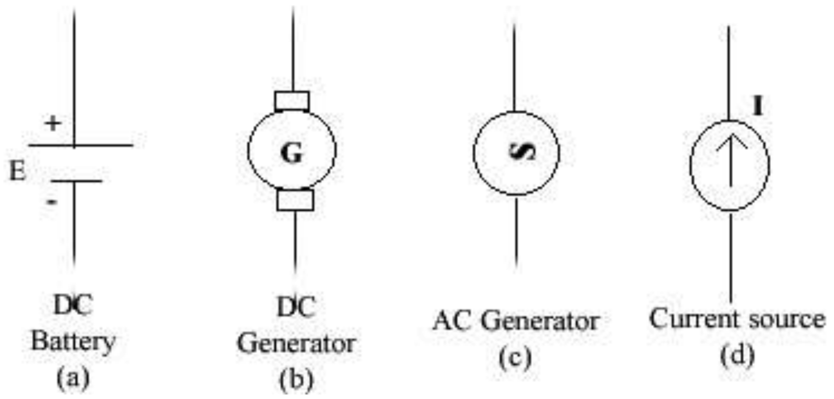
$P = 30$ watts.

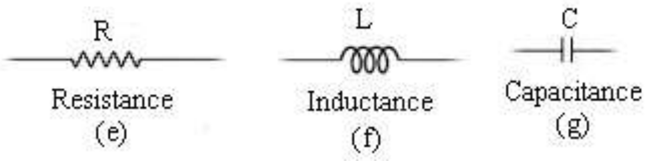
1.4 Electrical Network Circuit:

Any arrangement of electrical energy source (i.e. batteries, generator, etc.,) resistance and other circuit elements (e.g., inductance, capacitance) is called as electrical network.



Symbols of various Circuit Elements:





Active element: The elements which supply electrical energy (or) can able to generate electric energy is called active elements.
e.g., (a) to (d).

Passive element: Elements which consumes electrical energy or store energy is called passive elements.
e.g., (e) to (g).

1.5 Network Terminology

DC Circuit: The closed path followed by an electric current (DC) is called an electrical circuit. The essential parts of an electrical circuit are the source of power (battery, generator, etc.) the conductor to carry the current and the loads.

Load: The device that utilses the electrical energy, e.g., lamp, motor, heater, etc. Load may be a part of an electrical circuit (or) may not.

Parameters: The various elements of an electric circuit are called its parameter (i.e., resistance, capacitance, inductance, etc.). They may be in groups (or) separated in a circuit.

Node: Node is a junction where two or more circuit elements can connect together.

Minor node - Point at which two elements are *connected*.
Major node - Point at which three or more element are connected.

Branch: It is a part of a circuit which lies between two junction points.

Loop: A loop is any closed path of a circuit.

Mesh: A mesh is the most elementary form of a loop and cannot be further divided into other loops.

Problems

1.5 The resistance of a 230 V lamp is 270Ω. Calculate the current taken by the lamp.

Solution:
Given:

Resistance of the lamp, $R = 270\Omega$.
Voltage of the lamp, $V = 230V$.

To find:

Current of the lamp, I = ?
By Ohm's law, we know that

$$V = I \bullet R$$

$$\therefore I = \frac{V}{R} = \frac{230}{270} = 0.85 \text{ Amps.}$$

$\therefore$ The current taken by the lamp = 0.85 Amps.

1.6 An electric lamp consumes 100 watts of power. The supply Voltage is 230 V. Determine a) the current flowing through the filament b) its resistance c) the energy consumed in 45 minutes.

Solution:
Given:

Power consumed by electric lamp, $P = 100$ watts.

Supply Voltage, $V = 230\ V$.

Time, $T = 45$ minutes.

a) Current flowing,

$$I = \frac{P}{V} \text{ from } [P = V \cdot I]$$

$$I = \frac{100}{230}$$

$$\therefore I = 0.434 \text{ Amps.}$$

b) Resistance of filament,

$$R = \frac{V}{I} \text{ from } [P = V \cdot I]$$

$$\text{R}= \frac{230}{0.434}$$

$$\therefore R = 529.95\Omega.$$

c) Energy consumed in 45 min,

$$E = P \times T$$

$$= 100 \times \frac{45}{60}$$

$$E = 75 \text{ watts (or) } 0.07 \text{ kw.}$$

1.6 **Kirchoff's Law:**

Gustov Kirchoff a German scientist gave his findings with electrical circuit in a set of two laws.

1. Kirchoff's Current law or First law,

2. Kirchoff's Voltage law or Second law.

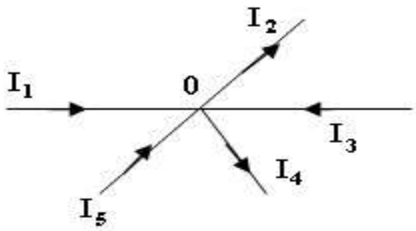
1.6.1 Kirchoff's Current law (KCL)

Statement:

"The algebraic sum of currents meeting a junction or node in an electrical circuit is zero".

$$\Sigma I = 0$$

Explanation and Velocity



Consider the figure shown. In KCL the sign of the quantity is taken into account. From the figure, there are five currents I_1, I_2, I_3, I_4 and I_5 meeting at a point or node "0".

Assume, the current flowing towards "0" is positive and the current flowing away from "0" is negative.

Applying Kirchoff's Current law at the junction "0" we get,

$$(+I_1)+(-I_2)+(+I_3)+(-I_4)+(+I_5)=0$$

$$I_1 - I_2 + I_3 - I_4 + I_5 = 0$$

$$I_1 + I_3 + I_5 = I_2 + I_4$$

From the above equation, the sum of incoming current is equal to the sum of the outgoing current.

Now the statement of KCL can also be written as "*the sum of the currents flowing towards any junction in an electrical circuit is equal to the sum of the current flowing away from the junction*".

Validity of KCL

KCL is true because electric current is merely flow of electrons and they cannot accumulate at any point in the circuit.

1.6.2 Kirchoff's Voltage Law

Statement:

"In a closed circuit the sum of the potential drops is equal to the sum of the potential rises"

(or)

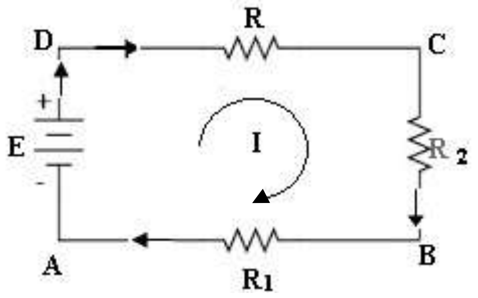
"In a closed circuit or mesh, the algebraic sum of all the emf's plus the algebraic sum of all the voltage drops is zero".

i.e., $\begin{bmatrix} \text{Algebraic Sum} \\ \text{of all emf} \end{bmatrix} + \begin{bmatrix} \text{Algebraic Sum} \\ \text{of all Voltage drops} \end{bmatrix} = 0.$

Mathematically,

$$\sum E + \sum V = 0.$$

Consider the below circuit,



Consider the entire loop ABCDA,

Sum of emf = E

the potential drop across AB, $V_{AB} = I . R_1$

the potential drop across BC, $V_{BC} = I . R_2$

the potential drop across CD, $V_{CD} = I . R_3$

$\therefore$ Sum of potential drops = $I . R_1 + I . R_2 + I . R_3$

By KVL,

$$\sum emf + \sum \text{ Voltage drop} = 0$$

$$E + V_{AB} + V_{BC} + V_{CD} = 0$$

$$i.e., E + IR_1 + IR_2 + IR_3 = 0$$

$$\therefore E = IR_1 + IR_2 + IR_{3.}$$

i.e., The KVL can also be stated as "In a closed circuit or mesh, the algebraic sum of all emfs is equal to the algebraic sum of all voltage drops".

While applying KVL, algebraic sum are involved. So it is necessary to assign proper signs to the emfs and to the voltage drops.

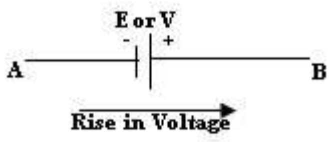
1.7 Sign Conversion:

Sign conversions are very important while considering Kirchoff's Voltage Law. It is very important to assign proper signs to emfs and voltage drop.

In order to assign proper signs both loop current and branch current direction are considered.

1.7.1 sign of emf:

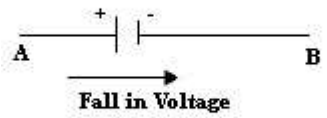
if we go from negative terminal of the battery or source to the positive terminal then there is rise in potential and so the emf should be given positive sign.



$\left[\begin{array}{c}\text{Emf is either}\\ \text{denoted as}\\ \text{E or V}\end{array}\right]$

Therefore + E (or) + V.

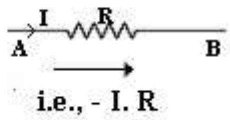
If we go from positive terminal of the battery (or) source to the negative terminal, then there is fall in potential and so the emf should be assigned negative sign.



$\therefore$ (– ve) = – E (or) – V.

1.7.2 Sign of Voltage drops:

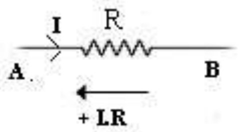
When the loop current and branch current are flowing in the same direction through a resistor then there is a fall in potential. So the sign of the voltage drop across the resistor is negative.



Fall in voltage or fall in potential.

When the loop current and branch current are flowing opposite to each other through a resistor then there is a rise in potential.

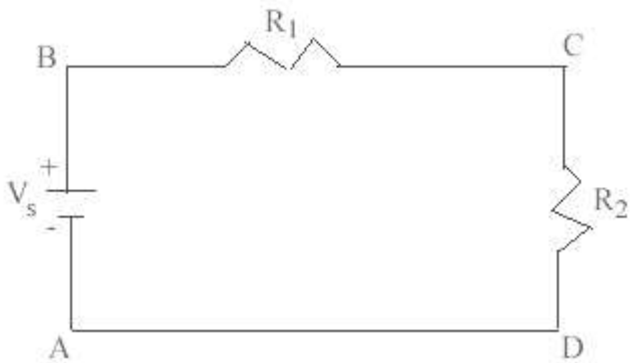
So the sign of the voltage drop across the resistor is positive.



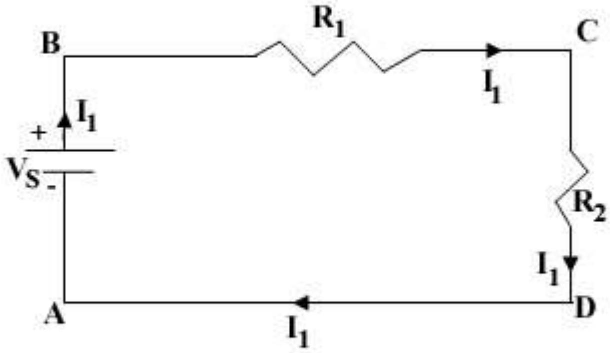
Rise in voltage or rise in potential.

Illustration – 1:

Using KVL solve the circuit below:-



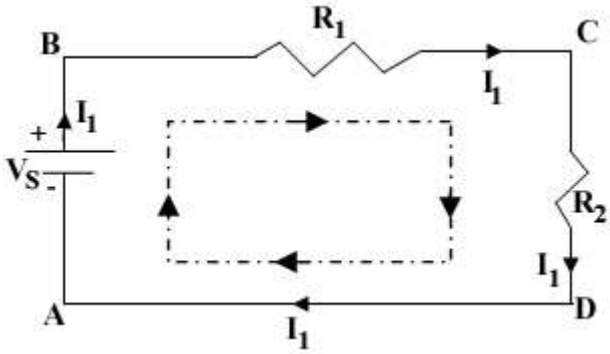
Step1:- Mark the branch current direction,



Note:-

Always the branch current will flow from the positive terminal of the battery or source.

Step2:- Mark the loop current direction,



Note:-

Always assume the loop current in clock wise direction.

Step 3:

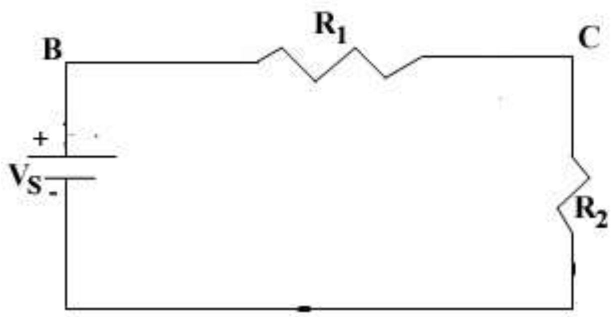
Apply KVL for ABCDA

$$V_s - I_1R_1 - I_1R_2 = 0$$

$$V_s - I_1(R_1 + R_2) = 0$$

$$\therefore V_s = I_1(R_1 + R_2)$$

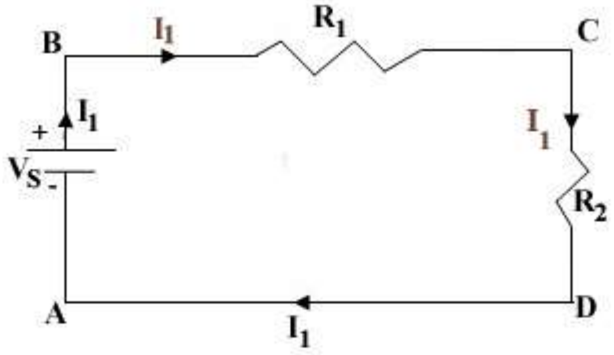
Illustration 2



Apply KVL for the above circuit and assume the loop current direction in anti-clockwise direction.

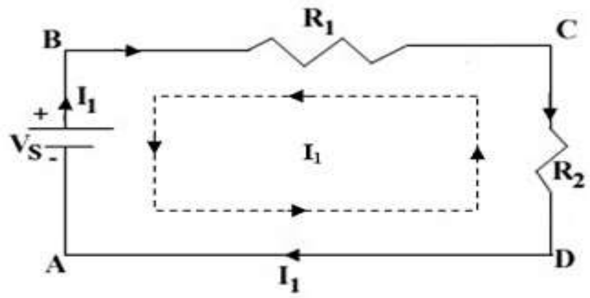
Step1

Mark the nodes as ABCDA, Mark the branch current direction.



Step2

Mark the loop current direction i.e., in anti-clockwise direction as given in questions.



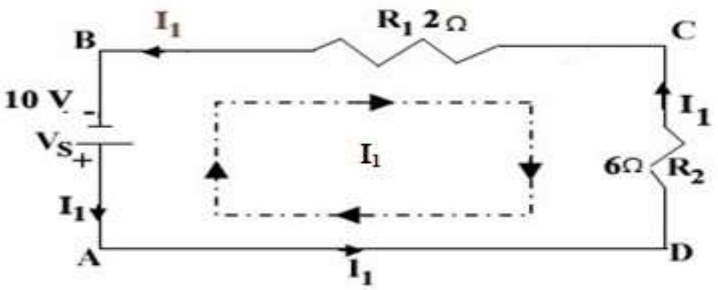
Step3

$$-V_s + I_1R_1 + I_1R_2 = 0$$
$$-V_s - I_1(R_1 + R_2) = 0$$
$$\therefore V_s = I_1(R_1 + R_2)$$

Simple problems on KVL and KCL

1.7 For the given circuit apply Kirchoff's law and find the current flowing through it?

Step1 Mark the nodes as ABCDA, Mark the branch and loop current direction.



Step2

Apply KVL,

$$-V_s + I_1R_1 + I_1R_2 = 0$$
$$-V_s + I_1(R_1 + R_2) = 0$$
$$-10 + I_1(2+6) = 0$$
$$-8I_1 = 10$$
$$\therefore I_1 = 1.25\text{Amps}$$

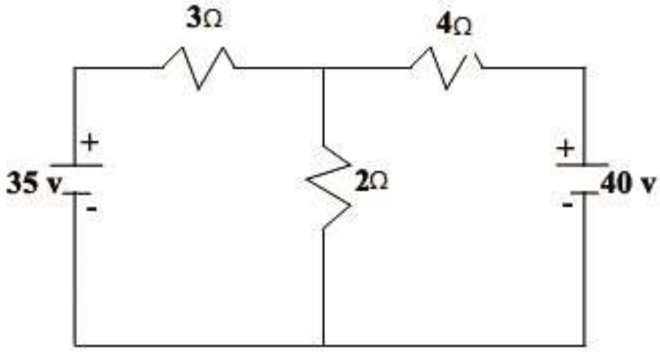
- Current through each element:-

$$I_{2\Omega} = I_{6\Omega} = I_1 = 1.25 \text{ Amps}$$

- Voltage drop across each element

$$V_{2\Omega} = I_{2\Omega}.R_{2\Omega} = 1.25 * 2 = 2.5\ V$$
$$V_{6\Omega} = I_{6\Omega}.R_{2\Omega} = 1.25 * 6 = 7.5\ V$$

1.8 Find the current through 2Ω resistor? Also find the current in each branch and current through 35 V and 40 V source [NOV - 2009]

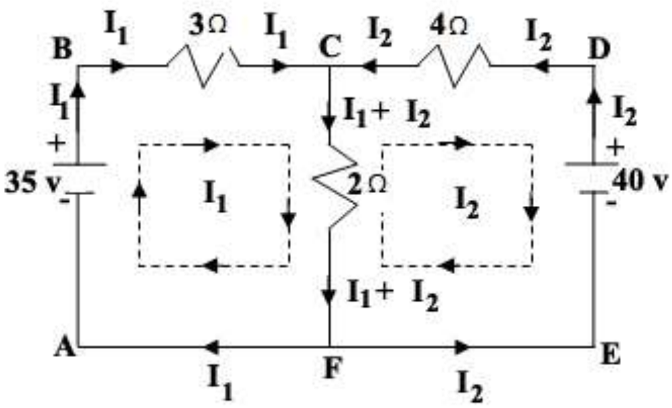


Solution

Step1

Mark the current as ABCDEFA.

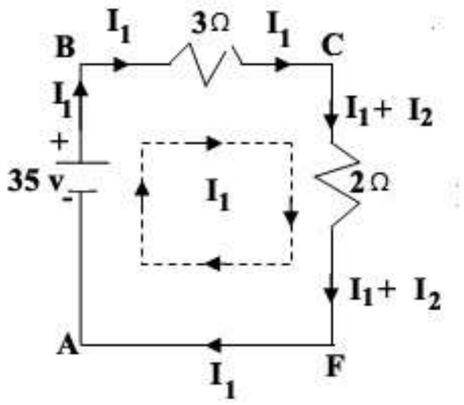
Mark the branch and loop current.



Step2

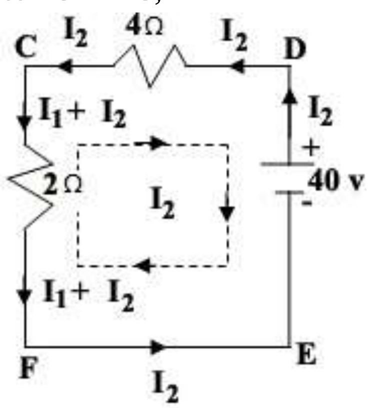
Since the above circuit has two loops (i.e), ABCFA & CDEFC we can apply KVL for each loop or mesh.

Apply KVL for the mesh ABCFA,



$$35 - 3I_1 - 2(I_1 + I_2) = 0$$
$$35 - 3I_1 - 2I_1 - 2I_2 = 0$$
$$35 - 5I_1 - 2I_2 = 0$$
$$-5I_1 - 2I_2 = -35$$
$$\therefore 5I_1 + 2I_2 = 35 \qquad - - - - 1$$

Apply KVL for the mesh CDEFC,



$$-40 + 4I_2 + 2(I_1 + I_2) = 0$$
$$-40 + 4I_2 + 2I_1 + 2I_2 = 0$$
$$-40 + 2I_1 + 6I_2 = 0$$
$$2I_1 + 6I_2 = 40 - - - - - - - - 2$$

Step3

Solving the equation 1 and 2 by matrix method

$$5I_1 + 2I_2 = 35 \qquad - - - - 1$$
$$2I_1 + 6I_2 = 40 - - - - - - - - 2$$
$$[R][I] = [V]$$
$$\begin{bmatrix} 5 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 35 \\ 40 \end{bmatrix}$$
$$\Delta = \begin{vmatrix} 5 & 2 \\ 2 & 6 \end{vmatrix} = 26$$
$$\Delta I_1 = \begin{vmatrix} 35 & 2 \\ 40 & 6 \end{vmatrix} = 130$$
$$\Delta I_2 = \begin{vmatrix} 5 & 35 \\ 2 & 40 \end{vmatrix} = 130$$

$$\therefore I_1 = \frac{\Delta I_1}{\Delta} = \frac{130}{26} = 5 \text{ A}$$
$$\therefore I_2 = \frac{\Delta I_2}{\Delta} = \frac{130}{26} = 5 \text{ A}$$

Step4

Current through 2Ω resistor,

$$I_{2\Omega} = I_1 + I_2$$
$$= 5 + 5$$
$$I_{2\Omega} = 10 \text{ Amps}$$

Step5

Current through each branch,

In branch

$$\text{AB} = I_1 = 5 \text{ A}$$
$$\text{BC} = I_2 = 5 \text{ A}$$
$$\text{CD} = I_2 = 5 \text{ A}$$
$$\text{DE} = I_2 = 5 \text{ A}$$
$$\text{EF} = I_2 = 5 \text{ A}$$
$$\text{FA} = I_1 = 5 \text{ A}$$
$$\text{CF} = I_1 + I_2 = 10 \text{ Amps}$$

Step6

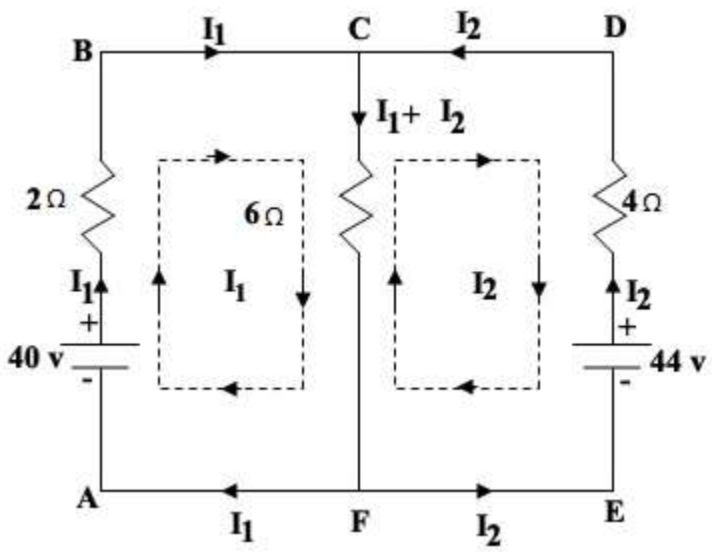
Current through the source 35V and 40 V

$$I_{35v} = I_1 = 5 \text{ A}$$
$$I_{40v} = I_2 = 5 \text{ A}$$

1.9 A battery of emf 40 V and the internal resistance of 2 Ω is connected in parallel with a second battery of 44 V and internal resistance of 4 Ω. A load resistance of 6 Ω is connected across the ends of the parallel circuit. Calculate the current in each battery and in the load? [April/May 2011; April 2010]

Solution

The circuit is,



Step1

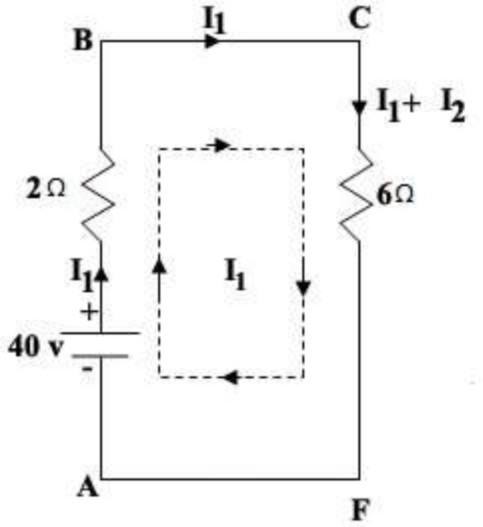
Mark the circuit as ABCDEFA,

Mark the branch and loop currents

[No. of Mesh = No. of Currents]

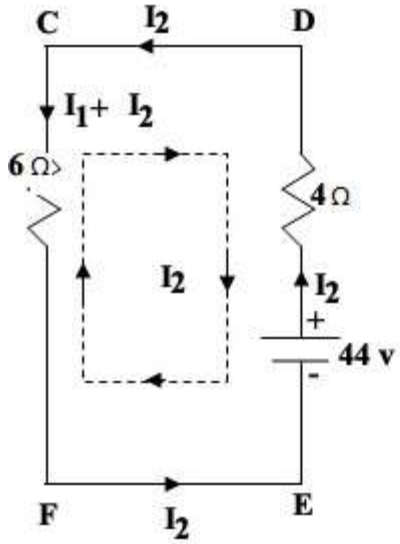
Step2

Apply KVL for the ABCFA,



$$40 - 2\,I_1 - 6\,(I_1 + I_2) = 0$$
$$40 - 2\,I_1 - 6I_1 - 6I_2 \ = 0$$
$$40 - 8\,I_1 - 6I_2 \ = 0$$
$$-8\,I_1 - 6I_2 \ = 0$$
$$\therefore 8I_1 + 6I_2 = 40 \qquad (1)$$

Apply KVL for CDEFC,



$$-44 + 4I_2 + 6(I_1 + I_2) = \ 0$$
$$-44 + 4I_2 + 6I_1 + 6I_2 = \ 0$$
$$6I_1 + 10\,I_2 = \ 44 \qquad (2)$$

Step3

Solve the equation 1 and 2 by matrix method and find the value of the Currents $I_1$and I_2.

$$\begin{bmatrix} 8 & 6 \\ 6 & 10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 40 \\ 44 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 8 & 6 \\ 6 & 10 \end{vmatrix} = 80 - 36 = 44$$

$$\Delta I_1 = \begin{vmatrix} 40 & 6 \\ 44 & 10 \end{vmatrix} = 400 - 264 = 136$$

$$\Delta I_2 = \begin{vmatrix} 8 & 40 \\ 6 & 44 \end{vmatrix} = 352 - 240 = 112$$

$$\therefore \Delta I_1 = \frac{\Delta I_1}{\Delta} = 3.09 \text{ A}$$

$$\therefore \Delta I_2 = \frac{\Delta I_2}{\Delta} = 2.54 \text{ A}$$

Step4

To find the current through 40 V and 44 V battery,

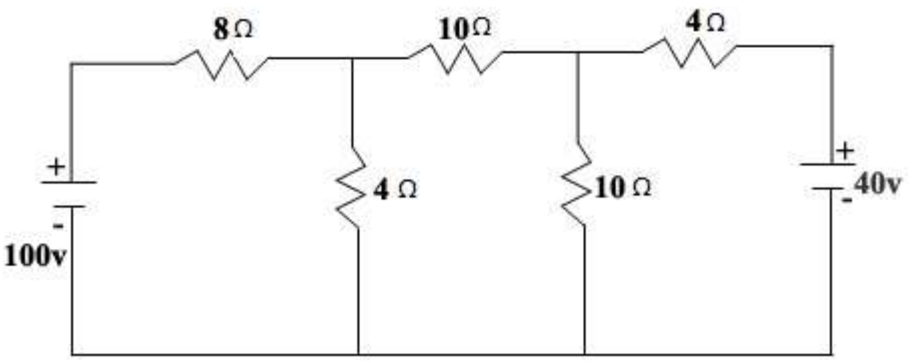
$$I_{40\text{v}} = I_1 = 3.09 \text{ A}$$
$$I_{44\text{v}} = I_2 = 2.54 \text{ A}$$

Step5

Current through the 6Ω load,

$$I_{6\Omega} = I_1 + I_2 = 3.09 + 2.54$$
$$\therefore I_{6\Omega} = 5.63 \text{ Amps}$$

1.10 Find the loop currents, current through each element and current across the voltage source, power delivered by 100 V and 40 V source. [April 2011; Nov 2011]



Step1

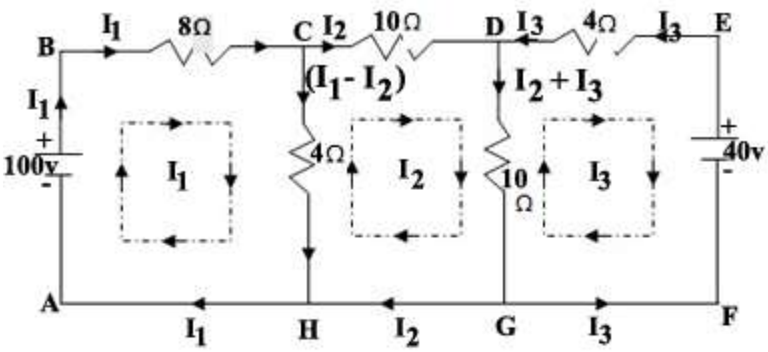
Mark the circuit as ABCDEFGHA,

Mark the loop currents and branch current directions.

No. of mesh = No. of currents

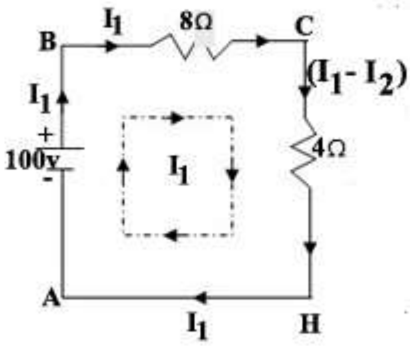
No. of mesh = 3 i.e., (ABCHA; CDGHC; DEFGD)

No. of currents = I_1, I_2 and I_3.



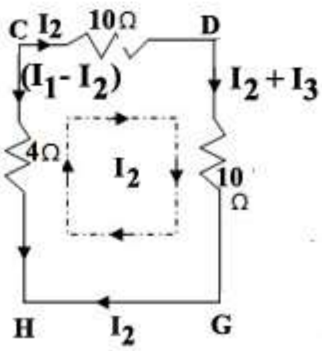
Step2

1) Apply KVL for the loop ABCHA,



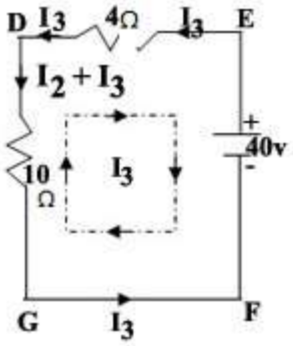
$$100 - 8I_1 - 4(I_1 - I_2) = 0$$
$$100 - 8I_1 - 4I_1 + 4I_2 = 0$$
$$100 - 12I_1 + 4I_2 = 0$$
$$-12I_1 + 4I_2 = -100$$
$$\therefore 12I_1 - 4I_2 = 100 \qquad (1)$$

2. Apply KVL for the loop ABCHA,



$$-10I_2 - 10(I_2 + I_3) + 4(I_1 - I_2) = 0$$
$$-10I_2 - 10I_2 - 10I_3 + 4I_1 - 4I_2 = 0$$
$$4I_1 - 24I_2 - 10I_3 = 0 \qquad (2)$$

3. Apply KVL for the loop DEFGD,



$$-40 + 4I_3 + 10(I_2 + I_3) = 0$$
$$-40 + 4I_3 + 10I_2 + 10I_3 = 0$$
$$10I_2 + 14I_3 = 40 \qquad (3)$$

Step3

Solve the equation 1, 2 and 3 by matrix method,

$$[R][I] = [V]$$

$$\begin{bmatrix} 12 & -4 & 0 \\ 4 & -24 & -10 \\ 0 & 10 & 14 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 100 \\ 0 \\ 40 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 12 & -4 & 0 \\ 4 & -24 & -10 \\ 0 & 10 & 14 \end{vmatrix} = -2608.$$

$$\Delta I_1 = \begin{vmatrix} 100 & -4 & 0 \\ 0 & -24 & -10 \\ 40 & 10 & 14 \end{vmatrix} = -2200.$$

$$\Delta I_2 = \begin{vmatrix} 12 & 100 & 0 \\ 4 & 0 & -10 \\ 0 & 40 & 14 \end{vmatrix} = -800$$

$$\Delta I_3 = \begin{vmatrix} 12 & -4 & 100 \\ 4 & -24 & 0 \\ 0 & 10 & 40 \end{vmatrix} = -6880$$

$$I_1 = \frac{\Delta I_1}{\Delta} = 8.4 \text{ A}$$

$$I_2 = \frac{\Delta I_2}{\Delta} = 0.306 \text{ A}$$

$$I_3 = \frac{\Delta I_3}{\Delta} = 2.63 \text{ A}$$

∴ Loop Currents are

$$I_1 = 8.4 \text{ A}$$
$$I_2 = 0.306 \text{ A}$$
$$I_3 = 2.63 \text{ A}$$

Step4

(i) Current through each element,

$$I_{8\Omega} = I_1 = 8.4 \text{ A}$$
$$I_{4\Omega} = I_1 - I_2 = 8.4 - 0.306 = 8.094 \text{ A}$$
$$I_{10\Omega} = I_2 = 0.306 \text{ A}$$
$$I_{10\Omega} = I_2 + I_3 = 0.306 + 2.63 = 2.936 \text{ A}$$

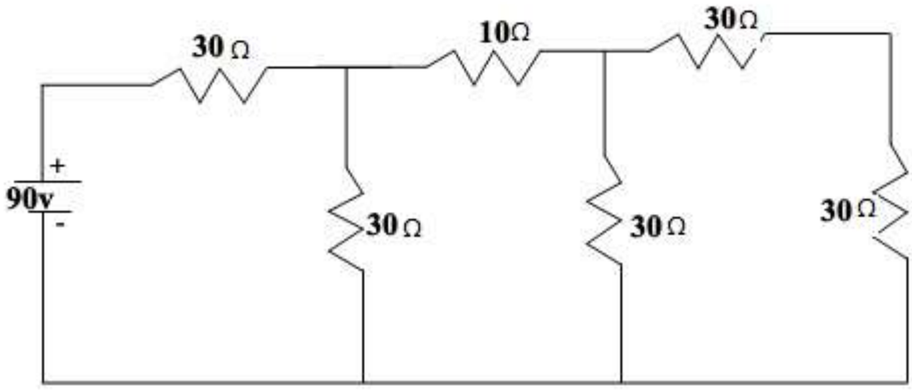
(ii) Current through the voltage sources

$$I_{100V} = I_1 = 8.4 \text{ A}$$
$$I_{40V} = I_3 = 2.36 \text{ A}$$

(iii) Power delivered by 100 V and 40 V source,

$$P_{100V} = V_1 * I_{100V} = 100 * 8.4 = 840 \text{ watts}$$
$$P_{40V} = V_2 * I_{40V} = 40 * 2.63 = 105.2 \text{ watts}$$

1.11 Find the current through each element and also find the current and power developed from the battery.



Solution:

Step1

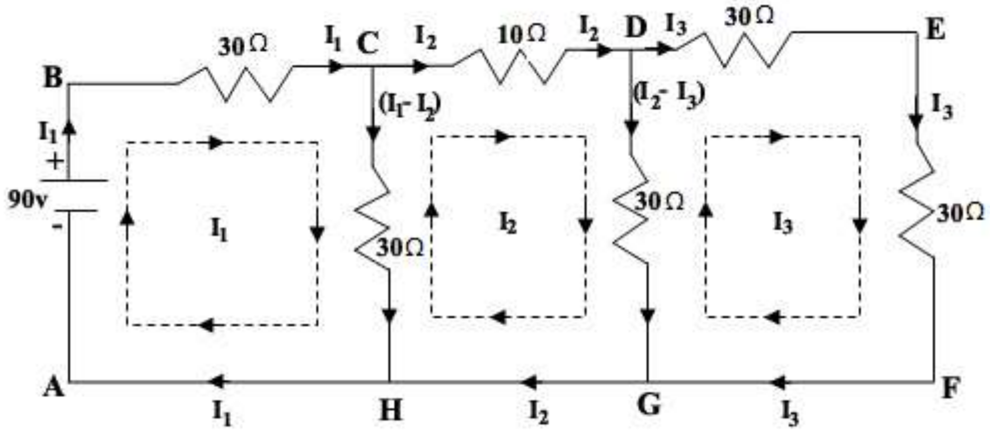
Mark the circuit or ABCDEFGHA,

Mark the loop and branch current directions

No. of mesh = 3

No. of currents = I_1, I_2 & I_3

i.e, No. of mesh = No. of Currents



Step2 Apply KVL,

(i) For mesh ABCHA,

$$90 - 30I_1 - 30\,(I_1 - I_2) = 0$$
$$90 - 30I_1 - 30I_1 + 30I_2 \;= 0$$
$$90 - 60I_1 + 30I_2 \;= 0$$
$$-60I_1 + 30I_2 \;= -90$$
$$\therefore\, 60I_1 - 30I_2 \;= 90 \qquad (1)$$

(ii) Apply KVL for CDGHC,

$$-10I_2 - 30(I_2 - I_3) + 30\,(I_1 - I_2) = 0$$
$$-10I_2 - 30I_2 + 30I_3 + 30I_1 - 30I_2 = 0$$
$$30I_1 - 70I_2 + 30I_3 = 0 \qquad (2)$$

(iii) Apply KVL for DEFGD,

$$-30I_3 - 30I_3 + 30(I_2 - I_3) = \;0$$
$$-30I_3 - 30I_3 + 30I_2 - 30\,I_3 = \;0$$
$$-90I_3 + 30I_2 = \;0$$
$$\therefore\, 30I_2 - 90I_3 = \;0 \qquad (3)$$

Step3

Find the currents I_1, I_2 & I_3 by using matrix,

$$[R][I] = [V]$$
$$\begin{bmatrix} 60 & -30 & 0 \\ 30 & -70 & 30 \\ 0 & 30 & -90 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 90 \\ 0 \\ 0 \end{bmatrix}$$
$$\Delta = \begin{vmatrix} 60 & -30 & 0 \\ 30 & -70 & 30 \\ 0 & 30 & -90 \end{vmatrix} = 243000$$
$$\Delta\, I_1 = \begin{vmatrix} 90 & -30 & 0 \\ 0 & -70 & 30 \\ 0 & 30 & -90 \end{vmatrix} = 486000$$
$$\Delta\, I_2 = \begin{vmatrix} 60 & 90 & 0 \\ 30 & 0 & 30 \\ 0 & 0 & -90 \end{vmatrix} = 243000$$
$$\Delta\, I_3 = \begin{vmatrix} 60 & -30 & 90 \\ 30 & -70 & 0 \\ 0 & 30 & 0 \end{vmatrix} = 81000$$
$$\therefore\; I_1 = \frac{\Delta I_1}{\Delta} = \frac{486000}{243000} = 2 \text{ Amps}$$
$$I_2 = \frac{\Delta I_2}{\Delta} = \frac{243000}{243000} = 1 \text{ Amps}$$
$$I_3 = \frac{\Delta I_3}{\Delta} = \frac{81000}{243000} = 0.333 \text{ Amps}$$

Step4

(i) Current supplied through the 90 V battery,

$$I_{90V} = I_1 = 2 \text{ Amps}$$

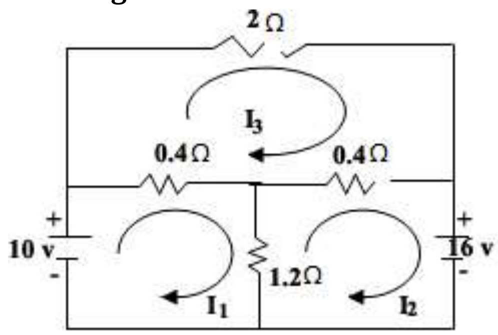
(ii) Power generated through the 90 V battery,

$$P_{90V} = V.I = 90 * 2 = 180 \text{ watts}$$

(iii) Current through each element,

$$I_{30\Omega} = I_1 = 2 \text{ Amps (BC Branch)}$$
$$I_{30\Omega} = I_1 - I_2 = 2 - 1 = 1 \text{ Amps (CH Branch)}$$
$$I_{10\Omega} = I_2 = 1 \text{ Amps (CD Branch)}$$
$$I_{30\Omega} = I_2 - I_3 = 1 - 0.333 = 0.667 \text{ Amps (DG Branch)}$$
$$I_{30\Omega} = I_3 = 0.333 \text{ Amps (DE Branch)}$$
$$I_{30\Omega} = I_3 = 0.333 \text{ Amps (EF Branch)}$$

1.12 Find the current through each element?

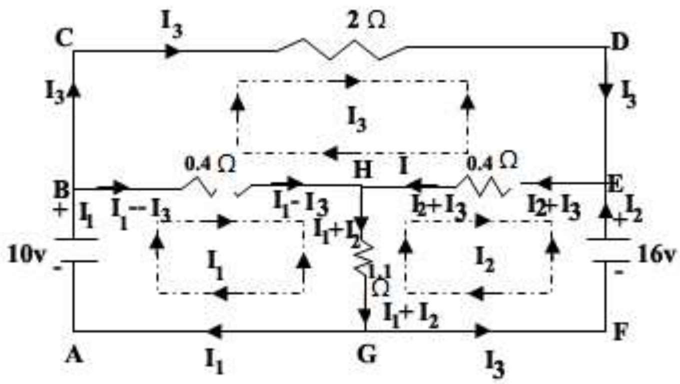


Solution

Step1

Name the circuit and draw the loop and branch current directions,

No. of loops = No. of currents = 3



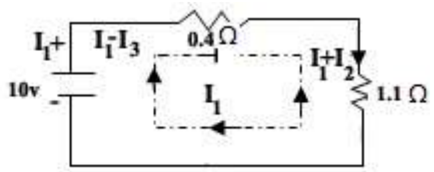
From the fig, in order to mark the current through the branch HG,

$$I_1 - I_3 + I_2 + I_3 = 0$$
$$I_1 + I_2 = 0$$

∴ The current through HG branch = $I_1 + I_2$.

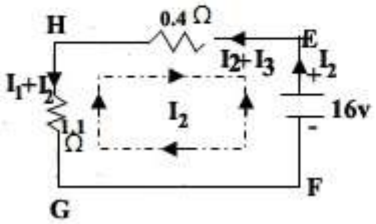
Step2 Apply KVL in all the 3 mesh,

1. Apply KVL in ABHGA,



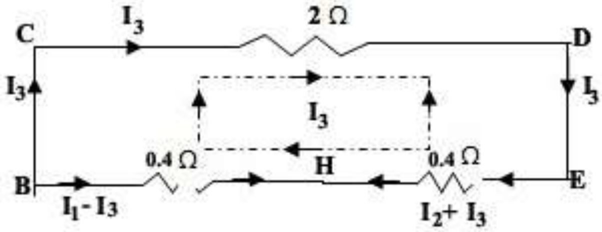
$$10 - 0.4\,(I_1 - I_3) - \ 1.1(I_1 + I_2) = 0$$
$$10 - 0.4\,I_1 + 0.4I_3 - \ 1.1I_1 - 1.1I_2 = 0$$
$$10 - 1.5\,I_1 - \ 1.1I_2 + 0.4I_3 = 0$$
$$-1.5\,I_1 - \ 1.1I_2 + 0.4I_3 = -\,10$$
$$1.5\,I_1 + \ 1.1I_2 - 0.4I_3 = 10 \qquad (1)$$

2. Apply KVL in HEFGH,



$$-16 + 0.4\,(I_2 + I_3) + 1.1(I_1 - I_3) = 0$$
$$-16 + 0.4I_2 + 0.4I_3 + 1.1I_1 + 1.1I_2 = 0$$
$$1.1I_1 + 1.5I_2 + 0.4I_3 = 16 \qquad (2)$$

3. Apply KVL in BCDEHB,



$$-2I_3 - 0.4(I_2 + I_3) + 0.4(I_1 - I_3) = 0$$
$$-2I_3 - 0.4I_2 - 0.4I_3 + 0.4I_1 - 0.4I_3 = 0$$
$$0.4I_1 - 0.4I_2 - 2.8I_3 = 0 \qquad (3)$$

Step3

Solving the equation 1, 2 and 3 by using matrix

$$1.5I_1 + 1.1I_2 - 0.4I_3 = 10 \qquad (1)$$
$$1.1I_1 + 1.5I_2 - 0.4I_3 = 16 \qquad (2)$$
$$0.4I_1 - 0.4I_2 - 2.8I_3 = 0 \qquad (3)$$
$$[R][I] = [V]$$
$$\begin{bmatrix} 1.5 & 1.1 & -0.4 \\ 1.1 & 1.5 & 0.4 \\ 0.4 & -0.4 & -2.8 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 16 \\ 0 \end{bmatrix}$$
$$\Delta = \begin{vmatrix} 1.5 & 1.1 & -0.4 \\ 1.1 & 1.5 & 0.4 \\ 0.4 & -0.4 & -2.8 \end{vmatrix} = -2.08$$

$$\Delta I_1 = \begin{vmatrix} 10 & 1.1 & -0.4 \\ 16 & 1.5 & 0.4 \\ 0 & -0.4 & -2.8 \end{vmatrix} = 11.44$$

$$\Delta I_2 = \begin{vmatrix} 1.5 & 10 & -0.4 \\ 1.1 & 16 & 0.4 \\ 0.4 & 0 & -2.8 \end{vmatrix} = -32.24$$

$$\Delta I_3 = \begin{vmatrix} 1.5 & 1.1 & 10 \\ 1.1 & 1.5 & 16 \\ 0.4 & -0.4 & 0 \end{vmatrix} = 6.24$$

$$\therefore\ I_1 = \frac{\Delta I_1}{\Delta} = \frac{11.44}{-2.08} = -5.5 \text{ Amps}$$

$$I_2 = \frac{\Delta I_2}{\Delta} = \frac{-32.24}{-2.08} = 15.5 \text{ Amps}$$

$$I_3 = \frac{\Delta I_3}{\Delta} = \frac{6.24}{-2.08} = -3 \text{ Amps}$$

Step4

Current through each element,

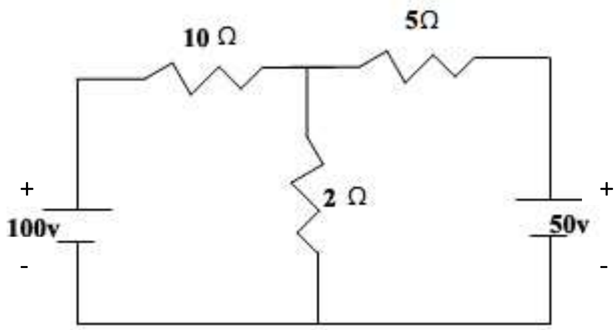
$$I_{0.4\Omega} = I_1 - I_3 = -5.5 — 3\ = -5.5 + 3 = \ -2.5 \text{ Amps}$$

$$I_{1.1\Omega} = I_1 +\ I_2\ = -5.5 - 15.5 = 10 \text{ Amps}$$

$$I_{0.4\Omega} = I_2 + I_3 = 15.5 - 3 = 12.5 \text{ Amps}$$

$$I_{2\Omega} = I_3\ = -3 \text{ Amps}$$

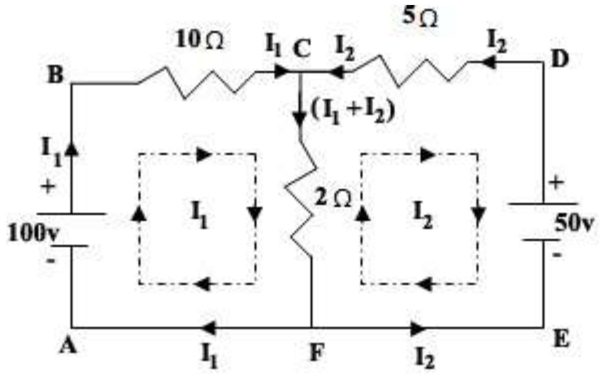
1.13 Find using Kirchoff's law, the voltage drop across the middle resistance?



Solution

Step1 Name the nodes or joints

Mark the branch and loop current directions.



No. of Mesh = No. of Current = 2 (i.e., I_1 and I_2)

Step 2

i. Apply KVL for ABCFA,

$$100 - 10I_1 - 2(I_1 + I_2) = 0$$
$$-10I_1 - 2I_1 - 2I_2 = -100$$
$$-12I_1 - 2I_2 = -100$$
$$\therefore 12I_1 + 2I_2 = 100 \qquad (1)$$

ii. Apply KVL for CDEFC,

$$-50 + 5I_2 + 2(I_1 + I_2) = 0$$
$$5I_2 + 2I_1 + 2I_2 = 50$$
$$2I_1 + 7I_2 = 50 \qquad (2)$$

Step 3

Solving equations 1 and 2 by matrix,

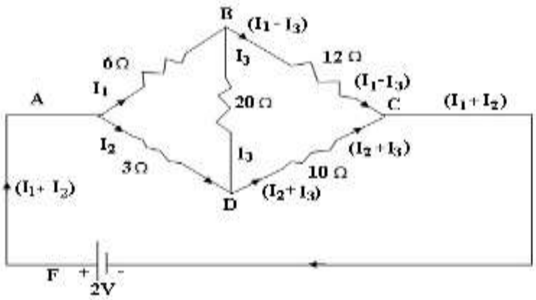
$$[R][I] = [V]$$
$$\begin{bmatrix} 12 & 2 \\ 2 & 7 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 100 \\ 50 \end{bmatrix}$$
$$\Delta = \begin{vmatrix} 12 & 2 \\ 2 & 7 \end{vmatrix} = 84 - 4 = 80$$
$$\Delta I_1 = \begin{vmatrix} 100 & 2 \\ 50 & 7 \end{vmatrix} = 700 - 100 = 600$$
$$\Delta I_2 = \begin{vmatrix} 12 & 100 \\ 2 & 50 \end{vmatrix} = 600 - 200 = 400$$
$$I_1 = \frac{600}{80} = 7.5 \text{ Amps}$$
$$I_2 = \frac{400}{80} = 5 \text{ Amps}$$

Step 4

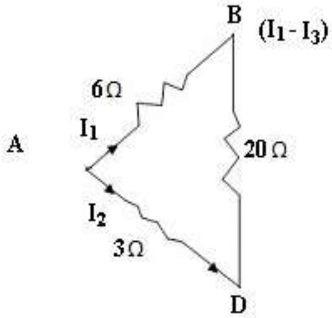
Current and voltage drop through middle resistance i.e., 2Ω

$$I_{2\Omega} = I_1 + I_2 = 7.5 + 5 = 12.5 \text{ Amps}$$
$$V_{2\Omega} = I_{2\Omega} \cdot R_{2\Omega} = 12.5 * 2 = 25 \text{ V.}$$

1.14 In the bridge network AB = 6Ω; BC = 12Ω; CD = 10Ω; DA = 3Ω; BD = 20 Ω and as an emf. of 2V. Calculate the I delivered to the battery?

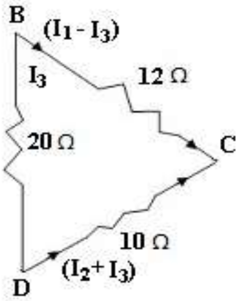


i) Apply KVL for loop ABDA,



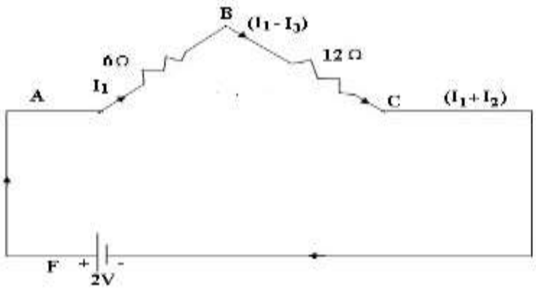
$$-6I_1 - 20I_3 + 3I_2 = 0$$
$$-6I_1 + 3I_2 - 20I_3 = 0 \qquad (1)$$

ii) Apply KVL for loop BCDB,



$$20I_3 - 12(I_1 - I_3) + 10(I_2 + I_3) = 0$$
$$20I_3 - 12I_1 + 12I_3 + 10I_2 + 10I_3 = 0$$
$$-12I_1 + 10I_2 + 42I_3 = 0 \qquad (2)$$

iii) Apply KVL for the loop ABCFA,



$$-6I_1 - 12(I_1 - I_3) + 2 = 0$$
$$-6I_1 - 12I_1 + 12I_3 + 2 = 0$$
$$-18I_1 + 12I_3 = -2 \qquad (3)$$

Solve the equation 1, 2 and 3 by matrix method,

$$\begin{bmatrix} -6 & 3 & -20 \\ -12 & 10 & 42 \\ -18 & 0 & 12 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} -6 & 3 & -20 \\ -12 & 0 & 42 \\ -18 & 0 & 12 \end{vmatrix} = -6156$$

$$\Delta I_1 = \begin{vmatrix} 0 & 3 & -20 \\ 0 & 10 & 42 \\ -2 & 0 & 12 \end{vmatrix} = -652$$

$$\Delta I_2 = \begin{vmatrix} -6 & 0 & -20 \\ -12 & 0 & 42 \\ -18 & 0 & 12 \end{vmatrix} = -984$$

1.8 Resistance in Series and Voltage Division Technique

The circuit in which resistances are connected end to end so that there is one path for the current flow is called series circuit.

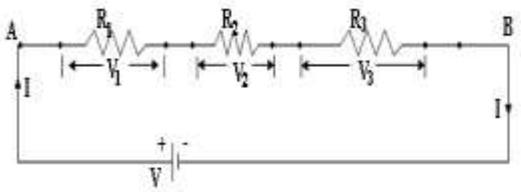


Fig. 1

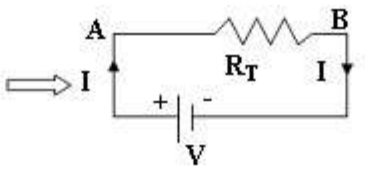


Fig. 2

In series circuit, the current through all the three resistors are same but the voltage drop across the each is different.

Let three resistances R_1, R_2 and R_3 are connected in series across a battery of V volts.

Let V_1, V_2 and V_3 be the voltage drop across R_1, R_2 and R_3.

By Ohm’s Law

$$\left.\begin{aligned} V_1 &= \text{Voltage drop across } R_1 = I \,.\, R_1 \\ V_2 &= \text{Voltage drop across } R_2 = I \,.\, R_2 \\ V_3 &= \text{Voltage drop across } R_3 = I \,.\, R_3 \end{aligned}\right\} \quad (1)$$

Therefore the total voltage drop across the circuit,

$$V = V_1 + V_2 + V_3 \qquad (2)$$

Sub. 1 in 2,

$$V = I \bullet R_1 + I \bullet R_2 + I \bullet R_3 \qquad (3)$$

By ohm’s law, the total voltage V,

$$V = I \bullet R_T \qquad (4)$$

Equate equation 3 and 4 we get,

$$I \bullet R_T = I \bullet R_1 + I \bullet R_2 + I \bullet R_3$$
$$I \bullet R_T = I(R_1 + R_2 + R_3)$$
$$\therefore R_T = R_1 + R_2 + R_3$$

Voltage Division Rule:

From the above circuit,

$$V = I \bullet R_T \qquad \text{(By Ohm's law)}$$
$$\text{i.e.,} \quad V = I(R_1 + R_2 + R_3)$$
$$\therefore I = \frac{V}{R_1 + R_2 + R_3} \qquad (1)$$

Voltage drop across the R_1,

$$V_1 = I \bullet R_1 \qquad (2)$$

Substitute equation 1 in 2

$$\therefore V_1 = \frac{V}{R_1 + R_2 + R_3} \bullet R_1$$

Similarly,

$$V_2 = \frac{V}{R_1 + R_2 + R_3} \bullet R_2$$
$$V_3 = \frac{V}{R_1 + R_2 + R_3} \bullet R_3$$

Therefore the equations can be written in words as,

$$\text{Voltage drop across a resistor} = \frac{\text{Applied Voltage}}{\text{Total Resistance}} * \text{ Corresponding Resistance}$$

Concept of Series circuit

1. The current flowing in all the parts of the circuit is same.
2. Voltage across the different elements will depend upon the resistance of elements.
3. Voltage drops are additive.
4. Resistances are additive.
5. Powers are additive.
6. Applied voltage equals to the sum of different voltage drops.

Disadvantages of Series circuit

1. If break occurs at any point in the circuit, no current will flow and the entire circuit becomes useless.
2. It is not applicable for practical lighting circuits, since the supply voltage increases depending upon the number of lamps (or) elements connected.
3. Since electrical devices have different current ratings, they are not connected in series for efficient operations.

1.9 Resistance in Parallel and Current Division Technique

Resistance in Parallel Combination

If one end of all the resistors is joined to a common point and the other ends are joined to another common point, the combination is said to be a parallel combination between two common points.

Let three resistances R_1, R_2 and R_3 are connected in parallel across a battery of V volts as shown below in fig (a).

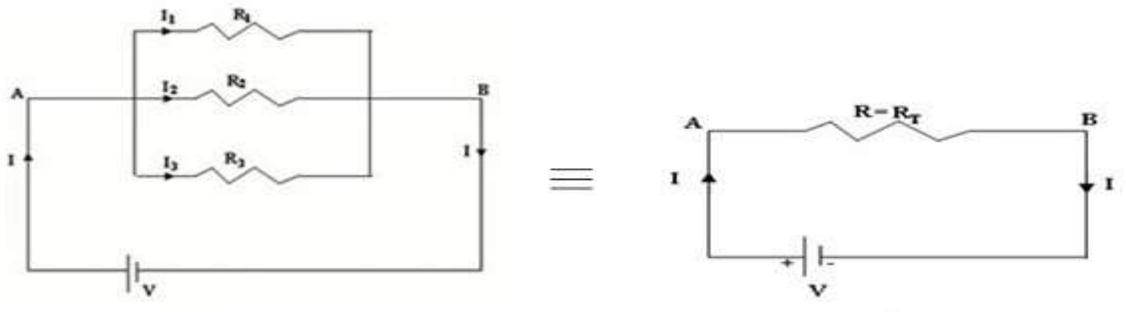


Fig. (a) Fig. (b)

The total current I divide into three parts,

I_1 flowing through R_1

I_2 flowing through R_2

I_3 flowing through R_3 and

The voltage drop across each resistance is same as that of supply voltage.

By ohm's law,

$$I_1 = \frac{V}{R_1}$$
$$I_2 = \frac{V}{R_2} \qquad (1)$$
$$I_3 = \frac{V}{R_3}$$

Total Current,

$$I = I_1 + I_2 + I_3 \qquad (2)$$

Substitute 1 in 2,

$$I = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$
$$I = V\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right]$$
$$\frac{I}{V} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \qquad (3)$$

By Ohm's law, from fig. (b),

$$\therefore \frac{I}{V} = \frac{1}{R_T} \qquad (4)$$

Equating equation 3 and 4, we get,

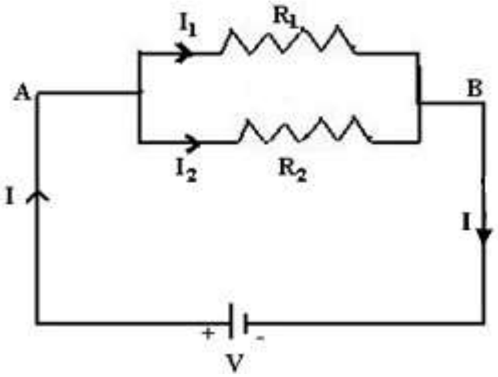
$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$\therefore$ The total resistance,

$$R_T = \frac{1}{\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right]}$$

Note

If there are two resistance R_1 & R_2 , then



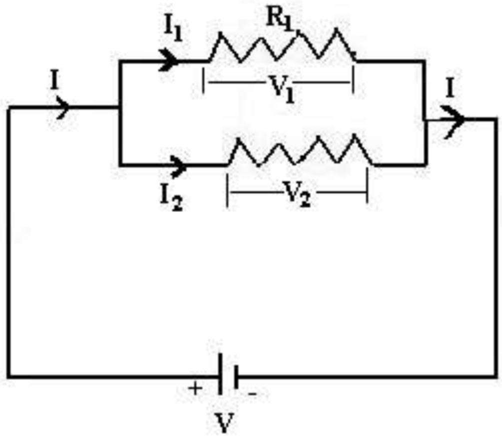
$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

$$= \frac{1}{\frac{R_1 + R_2}{R_1 R_2}}$$

$$\therefore R_T = \frac{R_1 R_2}{R_1 + R_2}$$

$$\therefore R_T = \frac{\text{Product of Resistances}}{\text{Sum of Resistance}}$$

Current Division Rule:



Supply voltage ≡ Voltage across R_1 = Voltage R_2

$$i.e., V \equiv I_1 R_1 = I_2 R_2$$

$$\therefore I_1 R_1 = I_2 R_2$$

$$\therefore I_2 = \frac{I_1 R_1}{R_2} \qquad (1)$$

Total Current, $I = I_1 + I_2$ (2)

Substitute equation 1 in 2,

$$\therefore I = I_1 + \frac{I_1 R_1}{R_2}$$

$$= \frac{I_1 R_2 + I_1 R_1}{R_2}$$

$$I = \frac{I_1 (R_1 + R_2)}{R_2} \qquad (3)$$

$\therefore$ Current through R_1,

$$I_1 = \frac{I.R_2}{R_1 + R_2}$$

$$i.e., I_1 = \frac{\text{Supply Current } * \text{ Opposite Resistance}}{\text{Sum of Resistance}}$$

Similarly,

$$I_2 = \frac{I.R_1}{R_1 + R_2}$$

Concepts of Parallel Circuits:

i. Voltage is same across all the elements.
ii. Current through the various resistors are different.
iii. Total resistances of a parallel circuit is always less than the smallest of the resistances.
iv. Powers are additive.
v. Source current is the sum of branch currents.
vi. Conductance's are additive.

Advantages of parallel circuits:

i. Electrical appliances of different power ratings may be rated for the same voltage. Example: In a house, tube lights, bulbs, fridge, fans, motor are all rated for 230 V. But their power ratings are different.

   In such cases, if they are all connected in parallel the performance of other devices will not be affected.

ii. In case a break occurs in any of the branch circuits, it will not affect the other branch circuits.

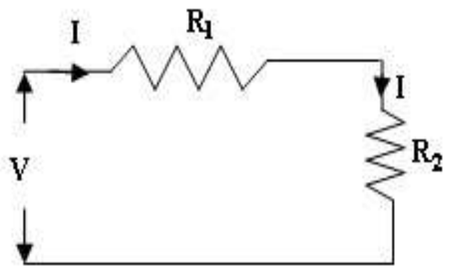
Comparison between series and parallel circuits

Series Circuit	**Parallel Circuit**
1. Current is same through all the elements.	Current is divided inversely proportional to resistance
2. Voltage is distributed. It is proportional to the resistance.	Voltage is same across each element in the parallel combination.
3. Total (or) equal resistance is equal to the sum of individual resistances.	Reciprocal of equal resistance is equal to the sum of reciprocals of individual

$i.e., R = R_1 + R_2 + R_3$	resistances. i.e., $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
4. There is only one path for flow of Current	There are more than one path for the flow of current

Illustration of how to solve series and Parallel circuits:

Illustration 1:

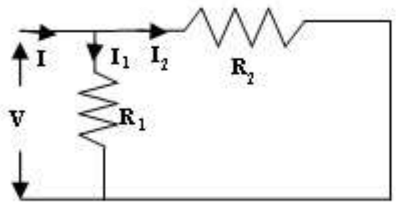


- The given circuit is a series circuit.
- The total resistance,

  $R_T = R_1 + R_2$

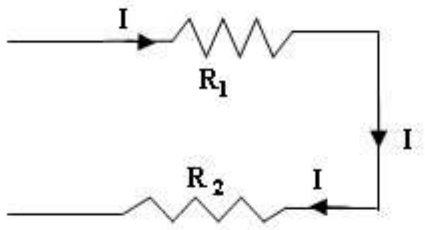
- Because R_1 and R_2 are connected in series.

Illustration 2:



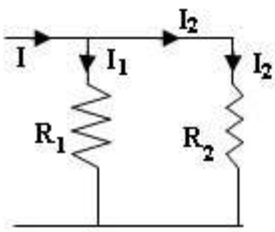
- It is a parallel circuit since R_1 is connected in parallel with R_2
- $R_T = \frac{R_1 R_2}{R_1 + R_2}$

Illustration 3:



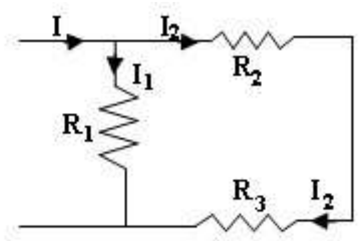
- It is a series circuit, since R_1 is in series with R_2
- $R_{T=} R_1 + R_2$

Illustration 4:



- It is a parallel circuit
- R_1 is in parallel with R_2
- $R_T = \frac{R_1 R_2}{R_1 + R_2}$

Illustration 5:

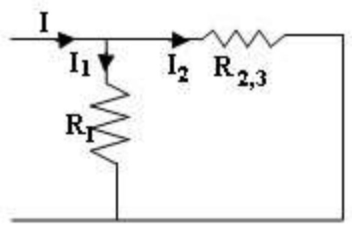


❖ It is a combination of parallel and series circuit. Hence we need to reduce the circuit in order to find the value of R_T.

Here R_2 is in series with R_3,

$$\therefore R_2, R_3 = R_2 + R_3$$

Therefore the above circuit can be reduced as,



❖ Here R_1 and $R_{2,3}$ are in parallel

$$\therefore R_T = \frac{R_1 \bullet R_{2,3}}{R_1 + R_{2,3}}$$

Therefore, the reduced circuit is,

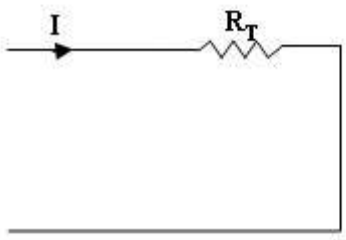
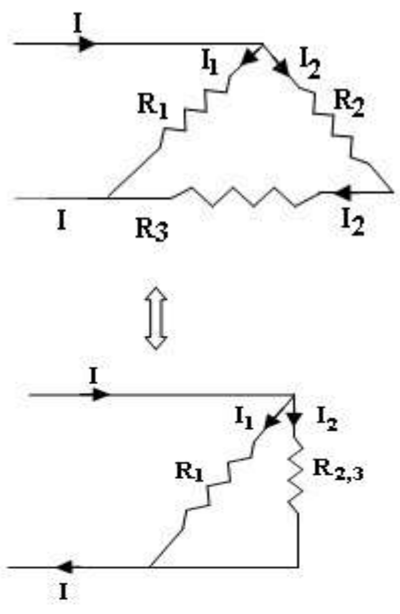


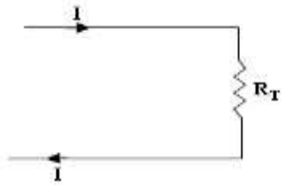
Illustration 6:



$R_{23} = R_2 + R_3$

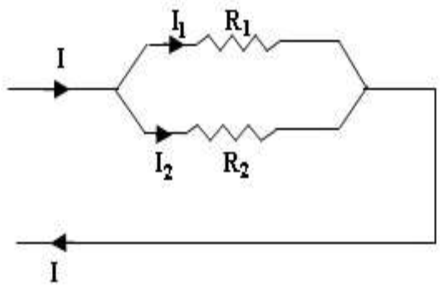
Since R_2 and R_3 are in series

The above circuit has two resistances R_1 and $R_{2,3}$, which are connected in parallel with each other.



$$R_T = \frac{R_1 \bullet R_{2,3}}{R_1 + R_{2,3}}$$

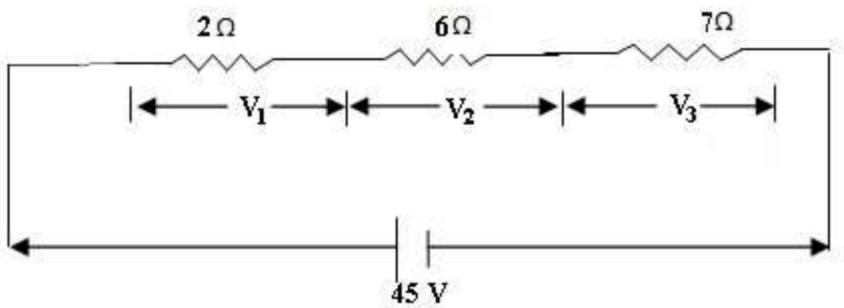
Illustration 7:



- It is a parallel circuit.
- $R_T = \dfrac{R_1 \bullet R_2}{R_1 + R_2}$

Problems on Series Circuit

1.15 In the circuit shown below, find the current, the voltage drop across each resistor and the power dissipated in resistor.



Given $R_1 = 2\Omega; R_2 = 6\Omega; R_3 = 7\Omega.$

Solution: Total Resistance,

$$R = R_1 + R_2 + R_3 \rightarrow R = 2 + 6 + 7 = 15\Omega$$

Supply voltage $= V = 45V$

Circuit current $I = \dfrac{V}{R} = \dfrac{45}{15} = 3\text{A}$

$$I = 3\text{A}$$

Voltage drop across 2Ω resistor $V_1 = IR_1 = 3 \times 2 = 6V$

Voltage drop across 6Ω resistor $V_2 = IR_2 = 3 \times 6 = 18V$

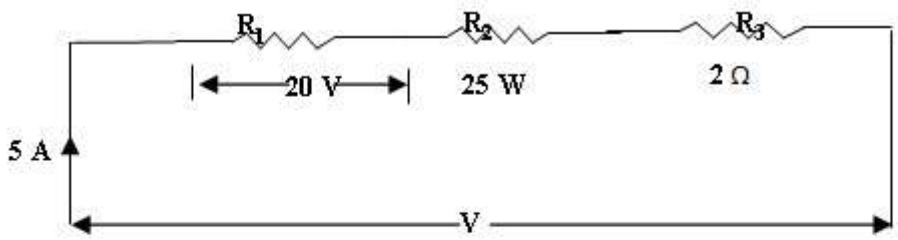
Voltage drop across 7Ω resistor $V_3 = IR_3 = 3 \times 7 = 21V$

Power dissipated in R_1 is $P_1 = I^2 R_1 = 3^2 \times 2 = 18\text{W}$

Power dissipated in R_2 is $P_2 = I^2 R_2 = 3^2 \times 6 = 54\text{W}$

Power dissipated in R_3 is $P_3 = I^2 R_3 = 3^2 \times 7 = 63\text{W}$

1.16 Three resistors R_1, R_2 and R_3 are connected in series across a constant voltage of "V" volts. The voltage drop across R_1 is 20 Volts. The power loss in R_2 is 25 Watts and R_3 has a resistance of 2Ω. Find the voltage "V" if the current is 5 Amps.



Solution

Voltage across R_1 is $V_1 = 20V$

Current, $I = 5A$

$$\therefore R_1 = \frac{V_1}{I} = \frac{20}{5}$$

$$R_1 = 4\Omega$$

Power loss in R_2 is $P_2 = 25W$

i.e., $P_2 = I^2 R_2$

$$R_2 = \frac{P_2}{I^2} = \frac{25}{5^2}$$

$$R_2 = 1\Omega$$

Total resistance, $R_T = R_1 + R_2 + R_3$

$$= 4 + 1 + 2$$

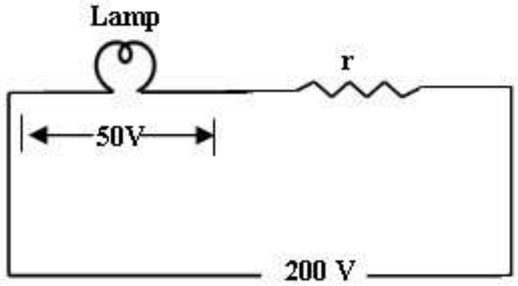
$$R_T = 7\Omega$$

Supply voltage, $V = IR$

$$= 5 \times 7$$

$$V = 35\text{Volts}$$

1.17 A lamp can work on 50 volt mains taking 2 amps. What value of resistance must be connected in series such that it can be operated from 200 V mains giving the same power.



Solution

Given

Voltage, $V = 50V$

Current, $I = 2\text{Amps}$

To find $r = ?$

Resistance of the lamp $R = \frac{V}{I}$

$$= \frac{50}{2} = 25\Omega$$

$$R = 25\Omega$$

Resistance connected in series with lamp $= r$

Supply voltage $= 200V$

Circuit current, $I = 2A$

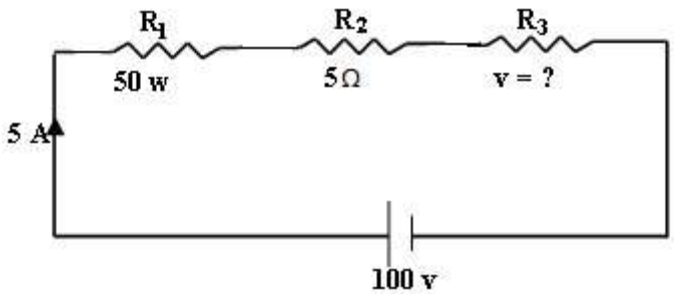
Total resistance, $R_T = \frac{V}{I} = \frac{200}{2} = 100\Omega$

i.e., $R_T = R + r$

$100 = 25 + r \Rightarrow r = 100 - 25$

$r = 75\Omega$

1.18 Three resistors R_1, R_2 and R_3 are connected in series. The power dissipated in P_1 is 50 w. The resistance of R_2 is 5Ω. The current through the series circuit is 5 A. When the supply voltage is 100 v, determine the power dissipated in circuit and voltage across R_3.



Solution

Given

$P_1 = 50w$

$R_2 = 5\Omega$

$I = 5A$

$V = 100\text{volts}$

To find, Power dissipated = ?

Voltage across R_3 = ?

$\therefore$ Total Power dissipated $= VI = I^2 R_T$

$$P = 100 \times 5 = 500w$$

Power Dissipated across R_1,

$$P_1 = I^2 R_1 = 50w \text{ (given)}$$

$$R_1 = \frac{50}{I^2} = \frac{50}{5^2} = \frac{50}{25}$$

$$R_1 = 2\Omega$$

By Ohm's law,

$$R_T = \frac{V}{I} = \frac{100}{5} = 20\Omega$$

$$R_T = 20\Omega$$

$$w.k.t.,\ R_T = R_1 + R_2 + R_3$$

$$\therefore R_3 = R_T - (R_1 + R_2)$$

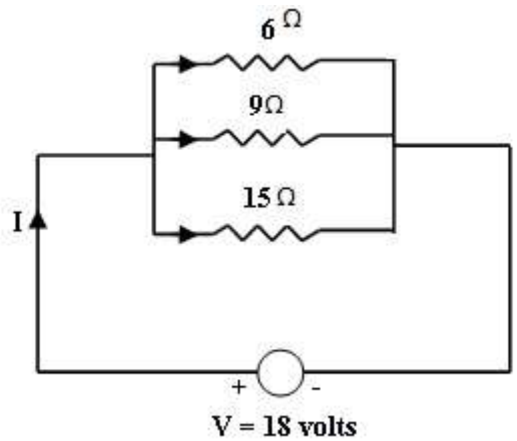
$$= 20 - (2+5) = 13\Omega$$

$$\therefore V_3 = IR_3 = 5(13) = 65\text{volts}$$

Problems on Parallel circuit

1.19 Three resistors of 6Ω, 9Ω and 15Ω are connected in parallel to a 18 v supply, calculate (a) the current in each branch of the network, (b) the supply current, (c) the total resistance of the network.

Solution



By Ohm's law, $I = \dfrac{V}{R}$

(i) Current through 6Ω resistor is $I_6 = \dfrac{V}{R_1} = \dfrac{18}{6} = 3\text{A}$ $\left[\dfrac{V}{R_1}\right]$

Current through 9Ω resistor is $I_9 = \dfrac{V}{R_2} = \dfrac{18}{9} = 2\text{A}$

Current through 15Ω resistor is $I_{15} = \dfrac{V}{R_2} = \dfrac{18}{15} = 1.2\text{A}$

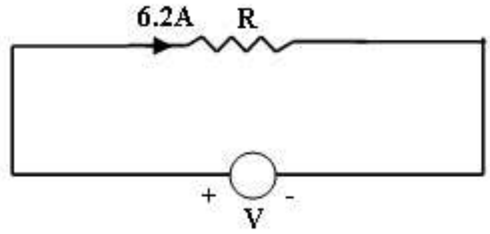
(ii) By KCL,

Supply Current = Source Current

$$I = 3 + 2 + 1.2$$

$$I = 6.2\text{A}$$

(iii) By Ohm's law,

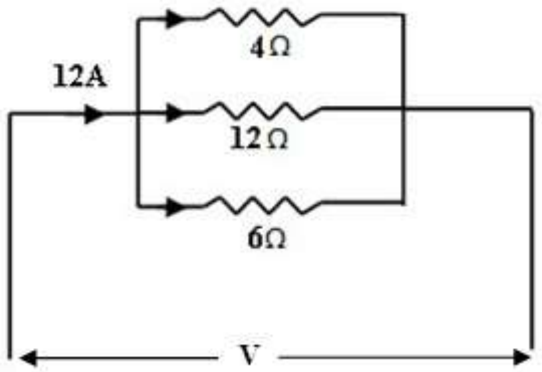


Total resistance,

$$R = \frac{V}{I} = \frac{18}{6.2}$$

$$\therefore R = 2.9\Omega$$

1.20 Three resistors 4Ω, 12Ω and 6Ω are connected in parallel. The total current is 12 A find the voltage across the parallel circuit and current across each branch.



$$\frac{1}{R_T} = \frac{1}{4} + \frac{1}{12} + \frac{1}{6}$$

$$= \frac{6}{12}$$

$$\therefore R_T = 2\Omega$$

Potential difference across the parallel circuit,

$$V = IR$$

$$= 12 \times 2$$

$$V = 24\text{volts}$$

Current through $4\Omega = I_1 = \frac{V}{R_1}$

$$= \frac{24}{4} = 6\text{A}$$

Current through $12\Omega = I_2 = \dfrac{V}{R_2}$

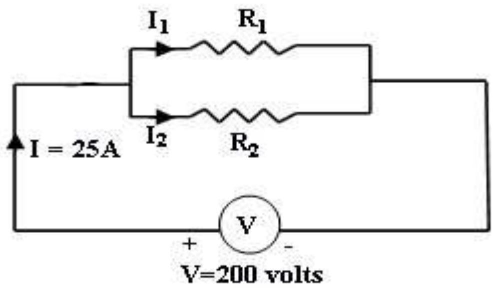
$$= \frac{24}{12} = 2\text{A}$$

Current through $6\Omega = I_3 = \dfrac{V}{R_3}$

$$= \frac{24}{6} = 4\text{A}$$

1.21 Two resistors are connected in parallel and a voltage of 200 volts is applied to the terminals. The total current taken is 25A and the power dissipated in one of the resistors is 1500 w. What is the resistance of each element.

Solution



Given

$V = 200$ Volts

$I = 25$ A

Power dissipated in resistor R_1

Say $P_1 = 1500$ watts

But $R_1 = \dfrac{V^2}{P_1}$ $\left[P = V^2/R\right]$

$$\Rightarrow R_1 = \frac{V^2}{P_1} = \frac{(200)^2}{1500}$$

$$= \frac{40000}{1500} = 26.67$$

$R_1 = 26.67\Omega$

Total resistor $R = \dfrac{V}{I} = \dfrac{200}{25} = 8\Omega$

W.K.T.,

$$R = \frac{R_1 R_2}{R_1 + R_2} = 8\Omega$$

$$\therefore 8 = \frac{R_1 R_2}{R_1 + R_2} \Rightarrow R_1 R_2 = 8(R_1 + R_2)$$

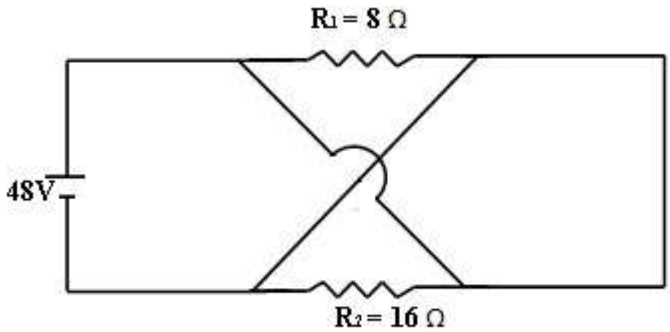
$$R_1R_2 = 8R_1 + 8R_2$$
$$R_1R_2 - 8R_2 = 8R_1$$
$$R_2\left[R_1 - 8\right] = 8R_1$$
$$R_2 = \frac{8R_1}{R_1 - 8} \qquad \because R_1 = 26.67\Omega$$
$$= \frac{8(26.67)}{26.67 - 8} = \frac{213.36}{18.67}$$
$$\therefore R_1 = 26.67\Omega$$
$$R_2 = 11.43\Omega$$

Note

R_2 Can be calculated by the following method also.

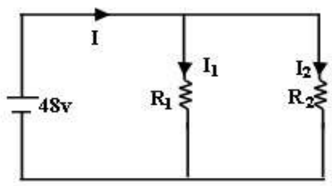
$$I_1 = \frac{V}{R} = \frac{200}{26.67} = 7.5A$$
$$\therefore I = I_1 + I_2$$
$$I_2 = I - I_1$$
$$= 25 - 7.5$$
$$I_2 = 17.5A$$
$$\therefore R_2 = \frac{V}{I_2} = \frac{200}{17.5} = 11.43\Omega$$
$$R_2 = 11.43\Omega$$

1.22 Calculate the current supplied by the battery in the given circuit shown below



Solution

For easy understanding the given circuit can be re-drawn as shown below.
There is no change in the connections.



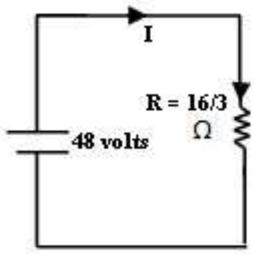
R_1 and R_2 are in parallel across the voltage of 48 volts.

The total resistance $R = \frac{R_1 R_2}{R_1 + R_2}$

$$= \frac{8 \times 16}{8 + 16}$$

$$R = \frac{16}{3} \Omega$$

$\therefore$ Equivalent circuit is



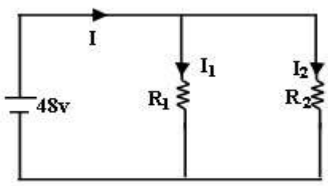
$\therefore$ Current from battery by Ohm's Law,

$$I = \frac{V}{R} = \frac{48}{\left(\frac{16}{3}\right)}$$

$$I = 9\text{A}$$

Note

From the circuit below, by applying ohm's law to R_1 and R_2, we get I_1 and I_2



$$I_1 = \frac{V}{R_1} = \frac{48}{8} = 6\text{A}$$

$$I_2 = \frac{V}{R_2} = \frac{48}{16} = 3\text{A}$$

$$\therefore I = I_1 + I_2$$

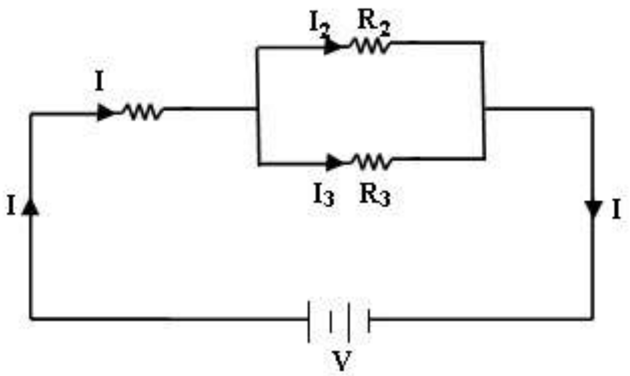
$$= 6 + 3 = 9\text{A}$$

$$I = 9\text{A}$$

1.11 Series-Parallel Circuits

As the name suggests, this circuit is a combination of series and parallel circuits. A simple example of such a circuit is shown in fig. below. Note that R_1 and R_2 are connected in parallel with each other and that is together connected in series with R_1.

One simple rule to solve such circuits is to first reduce the parallel branches to an equivalent series branch and then solve the circuit as a simple series circuit.



Referring to the above series-parallel circuit,

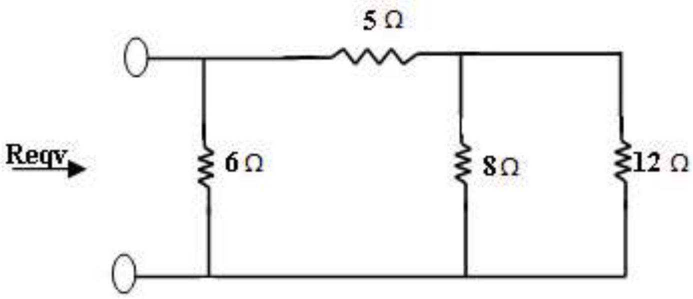
R_P for parallel combination $= R_1 + \dfrac{R_2 R_3}{R_2 + R_3}$

Total circuit resistance, $= R_1 + \dfrac{R_2 R_3}{R_2 + R_3}$

Voltage across parallel combination $= I + \dfrac{R_2 R_3}{R_2 + R_3}$

Problems on Series-Parallel Combination

1.23 Find the equivalent resistance of the circuit shown below:

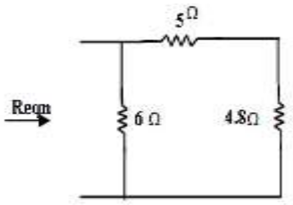


Solution

First reduce the last parallel combination $8\Omega \parallel 12\Omega$.

$$= \frac{8 \times 12}{8+12} = \frac{96}{20} = 4.8$$

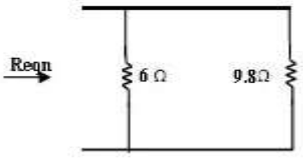
$\therefore$ The circuit becomes



5Ω and 4.8Ω are in series.

$\Rightarrow 5 + 4.8 = 9.8\Omega$ [Eqn. Series Combination]

$\therefore$ The circuit becomes

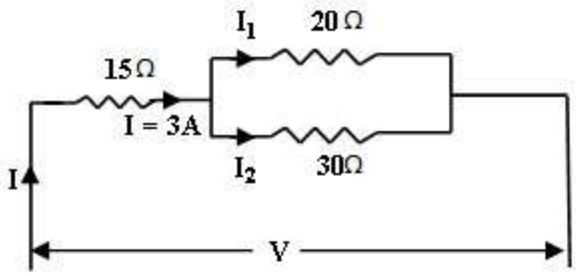


Again the parallel combination is reduced as

$$\frac{6\times 9.8}{6+9.8}=\frac{58.8}{15.8}=3.7\Omega$$

$\therefore$ R eqn = 3.7Ω

1.24 A circuit consists of 2 parallel resistors, having resistances of 20 Ω and 30Ω respectively, connected in series with 15Ω. If current through resistor is 3 A, Find i) the current through 20 Ω and 30Ω resistor, ii) the voltage across the whole circuit, iii) total power.



Solution:

(i) The total current 3A is divided between the 20 Ω and 30Ω resistors as shown below.

Current through 20 Ω, $I_1 = 3\times\frac{30}{(20+30)}$

$$I_1 = 1.8\text{A}$$

Current through 30 Ω, $I_2 = 3\times\frac{20}{(20+30)}$

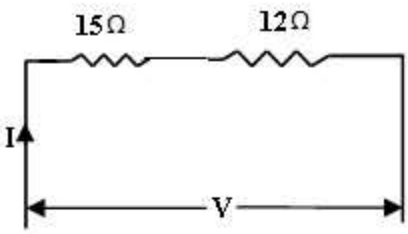
$$I_2 = 1.2\text{A}$$

(ii) For parallel circuit,

$$R_P = \frac{20\times 30}{20+30}=\frac{600}{50}$$

$$R_P = 12\Omega$$

$\therefore$ The circuit becomes



Total resistance $=15+12$

R eqn = $R_T = 27\Omega$

$\therefore$ Supply voltage $V = 3\times 27$

$= 81\text{V}$

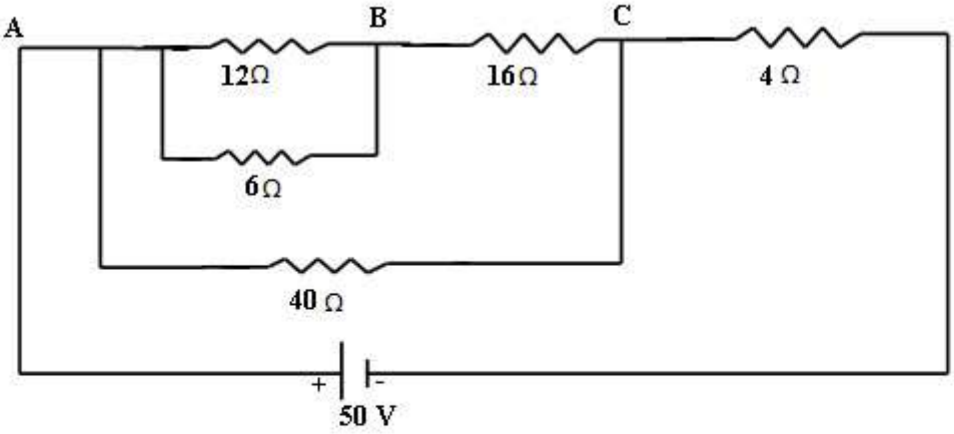
(iii) Total power

$= VI$ or I^2R

$= 81\times 3$

$= 243$ watts

1.25 Calculate the equivalent resistance of the following combination of resistors and also the source current



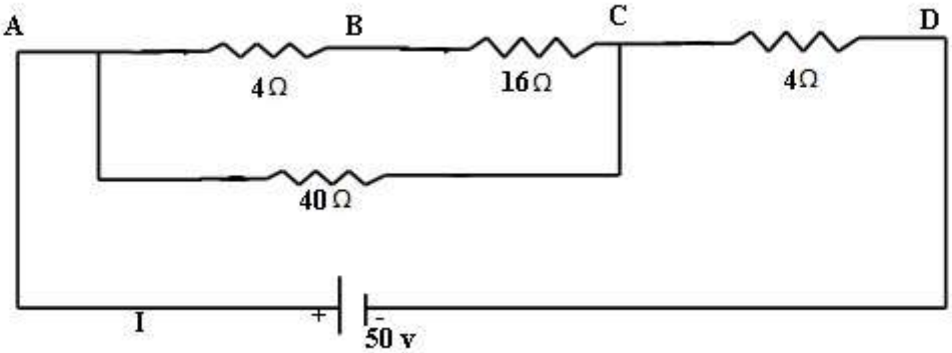
Solution

Start simplifying from inner combination

Step1: Replace parallel combination of 12 and 6 Ω to get the following circuit

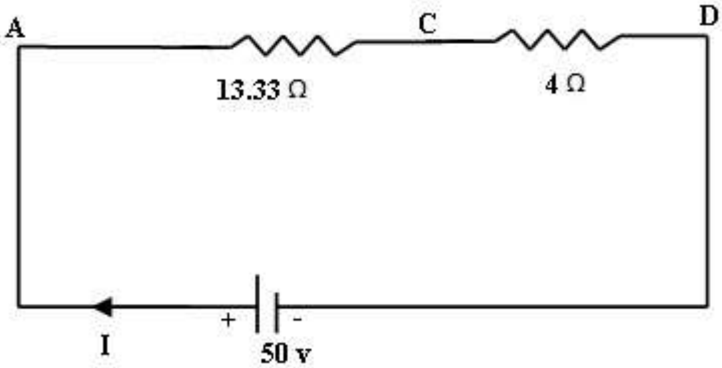
$$12 \| 16 = 12\Omega \text{ parallel with } 16\Omega$$

$$\therefore \frac{12\times 6}{12+6} = \frac{72}{18} = 4\Omega$$



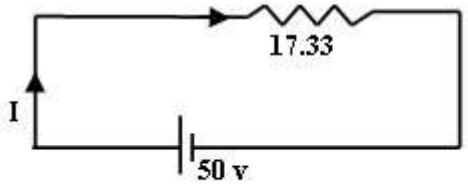
Step2: (4 + 16) and 40 are in parallel

Their equivalent resistance is $\frac{20\times 40}{20+40} = \frac{800}{60} = 13.3\Omega$



Step3: 13.33Ω and 4Ω are in series,

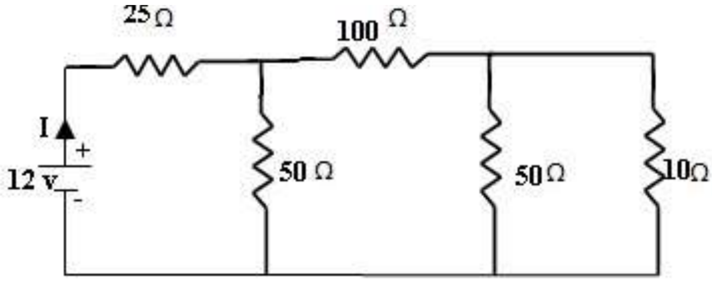
$$\therefore R_T = 13.33 + 4$$
$$R_T = 17.33\Omega$$



Step4: $I =$ Source Current

$$= \frac{V}{R_T} = \frac{50}{17.33} = 2.885\text{A}$$
$$I = 2.885\text{A}$$

1.26 Find the R equivalent and current flowing due to a source of 12v for the circuit shown.

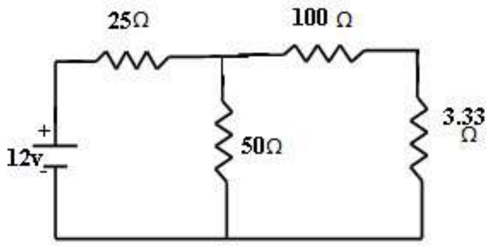


Solution

By reducing the last parallel combination

$$R_P = \frac{5\times10}{5+10} = \frac{50}{15} = 3.33$$
$$R_P = 3.33\Omega$$

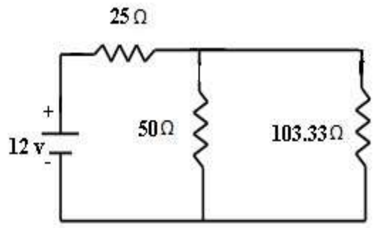
The circuit becomes,



Now, 100 Ω and 3.33Ω are in series,

$$\therefore 100 + 3.33$$
$$= 103.33\Omega$$

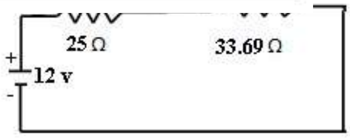
Then the circuit becomes



This parallel combination can be reduced by

$$= \frac{50 \times 103.33}{(50+103.33)} \qquad \left[\because 50 \,\|^{el}\, 103.33\right]$$

$$= 33.69\Omega$$

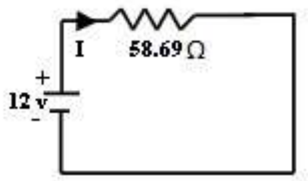


The two resistances are in series.

$$\therefore R_s = R_1 + R_2$$

$$= 25 + 33.69$$

$$\text{Req} = R_s = 58.69\Omega$$

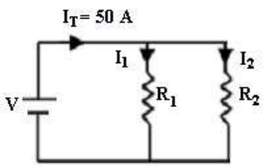


$$\therefore \text{Current } I = \frac{V}{R} = \frac{12}{58.69} = 0.204\text{A}$$

$$I = 0.204\text{A}$$

1.27 Two resistors R_1 and R_2 are connected in parallel with total resistance of I_t. Obtain the expression for the currents in the individual resistor. If $R_1 = 1.95\Omega$, $R_2 = 0.05\Omega$ and $I_t = 50\text{A}$, Find I_1 and I_2

Solution



Give data

$$I_T = 50\text{A}$$

$$R_1 = 1.95\Omega$$

$$R_2 = 0.05\Omega$$

To find I_1 and I_2

Total current $I_t = I_1 + I_2$

$$I_1 = I_t \times \frac{R_2}{R_1 + R_2}$$

$$= 50 \times \left[\frac{0.05}{(1.95+0.05)}\right]$$

$$= 50\times\left[\frac{0.05}{2}\right]$$

$$= 50\times 0.025$$

$$I_1 = 1.25$$

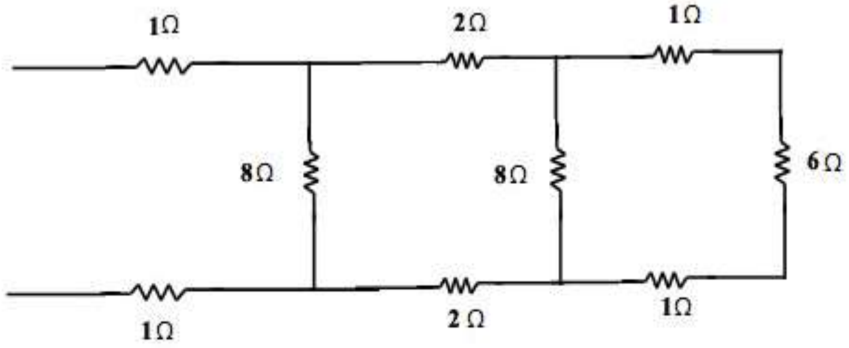
$$I_2 = I_T \times \frac{R_1}{R_1 + R_2}$$

$$= 50\times\left[\frac{1.95}{(1.95+0.05)}\right]$$

$$= 50\times 0.975 = 48.75\text{A}$$

$$I_2 = 48.75\text{A}$$

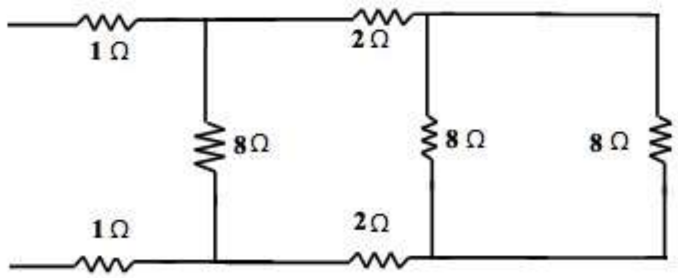
1.28



What is the equivalent resistance of the network.

Solution

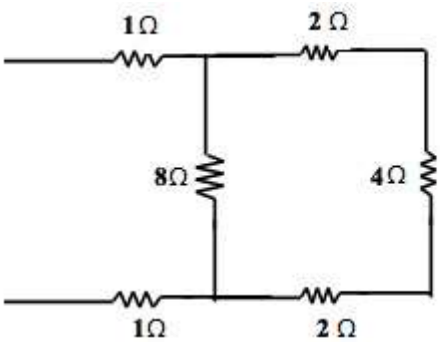
Since the last three resistances (1Ω, 6Ω and 1Ω) one series, we can redraw the circuit as shown by adding the 3 resistances.



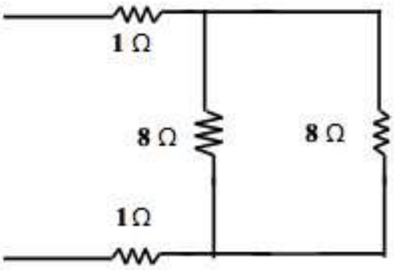
Now the two 8Ω are in parallel, Hence, the equivalent resistance of the parallel combination is

$$\therefore R_P = \frac{8\times 8}{8+8} = \frac{64}{16}$$

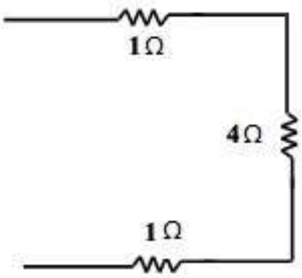
$$= 4\Omega$$



Now the resistors 2Ω, 4Ω and 2Ω are in series



$$R_P = \frac{8 \times 8}{8+8} = 4\Omega$$

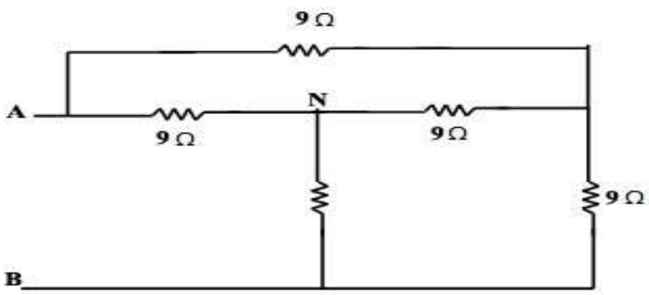


$\therefore$ Total Resistance = 1 + 4 + 1

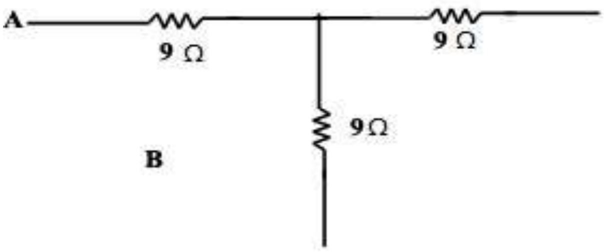
= 6 Ω

Problems on Series and Parallel Circuits

1.29 Find the resistance across AB for the network



Solution



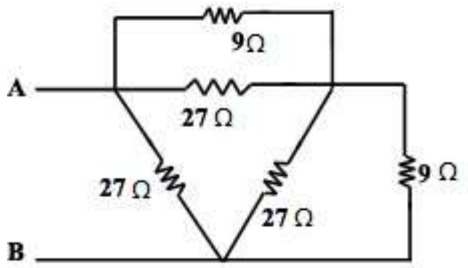
$$R_{AB} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_C} = \frac{(9\times 9)+(9\times 9)+(9\times 9)}{9}$$

$$R_{AB} = 27\Omega$$

$$R_{BC} = 27\Omega$$

$$R_{CA} = 27\Omega$$

Thus connecting star connection to delta

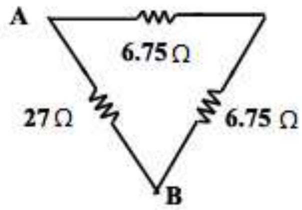


$27\Omega \,||\, 9\Omega$

$$R_P = \frac{R_1 R_2}{R_1 + R_2}$$

$$= \frac{27\times 9}{27+9}$$

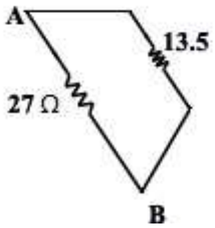
$$R_P = 6.75\Omega$$



$$R_S = R_1 + R_2$$

= 6.75+6.75

=13.5

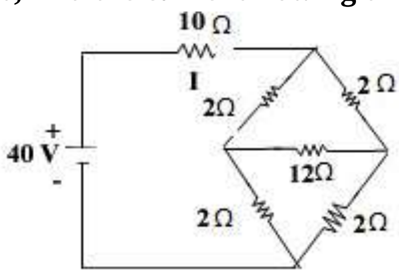


$27\Omega \,||\, 13.5\Omega$

$$R_{AB} = \frac{27\times 13.5}{27+13.5} = \frac{364.5}{40.5} = 9\Omega$$

$\therefore$ Resistance across AB for the network given is 9Ω.

1.30 For the circuit shown below, find the current flowing through the 10 Ω resistor.



Solution:

By applying Δ-Y transformation, the circuit is modified as

$$R_A = \frac{R_{AB} R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

$$= \frac{2 \times 2}{16} = \frac{4}{16} = \frac{1}{4}\Omega$$

$$R_A = \frac{1}{4}\Omega$$

$$R_B = \frac{R_{CA}R_{AB}}{R_{AB} + R_{BC} + R_{CA}}$$

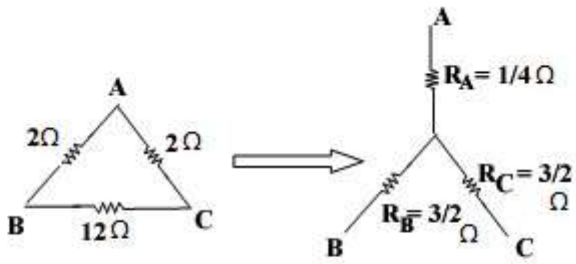
$$= \frac{2 \times 12}{16} = \frac{24}{16} = \frac{3}{2}\Omega$$

$$R_B = \frac{3}{2}\Omega$$

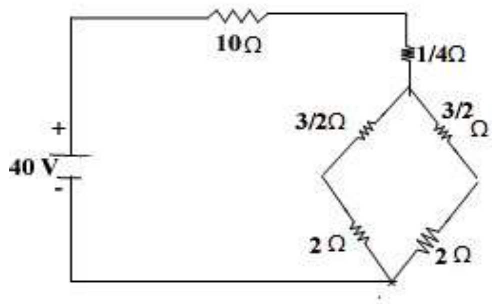
$$R_C = \frac{R_{CA}R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$

$$= \frac{2 \times 12}{16} = \frac{24}{16} = \frac{3}{2}\Omega$$

$$R_C = \frac{3}{2}\Omega$$



∴ The circuit becomes,



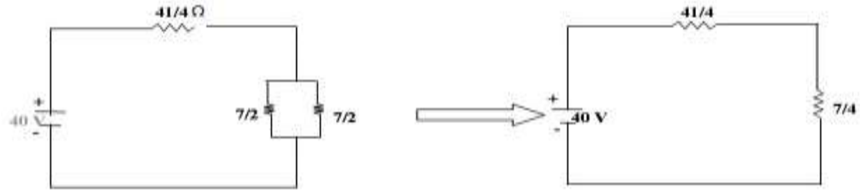
10Ω and $^1/_4$ Ω resistors one in series

$$10 + \frac{1}{4} = \frac{40 + 1}{4}$$

$$= \frac{41}{4}$$

Similarly $^1/_4\,\Omega$ and 2Ω are in series

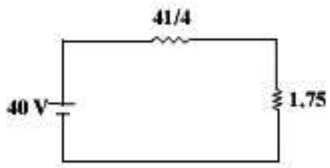
$$\frac{3}{2}+2=\frac{7}{2}\Omega$$



$^7/_2\,\Omega$ and $^7/_2$ are in parallel

$$=\frac{\frac{7}{2}\times\frac{7}{2}}{\frac{7}{2}+\frac{7}{2}}=\frac{\frac{49}{4}}{\frac{14}{2}}=\frac{49}{4}\times\frac{2}{14}=\frac{49}{28}=\frac{7}{4}$$

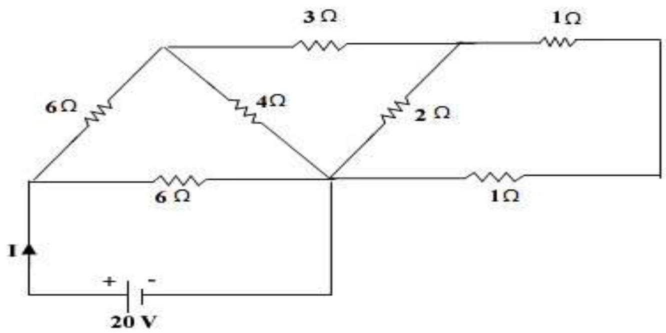
Req = $\frac{40}{4}+\frac{7}{4}=\frac{47}{4}=11.75\Omega$



$$I=\frac{V}{R}$$

$$I=\frac{40}{11.75}=3.4A$$

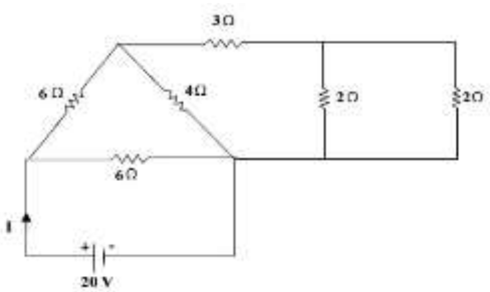
1.31 Find the current supplied by the source in the circuit shown in the figure below. Also determine the power in each resistance.



Solution:

The circuit diagram is simplified in the following steps.

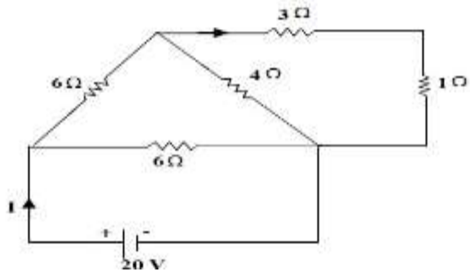
Step1:-



$$1\Omega+1\Omega$$

$$=2\Omega$$

Step2:-



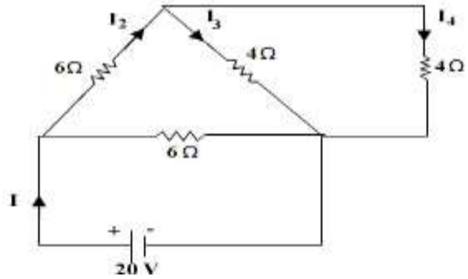
The last 2 resistors are in parallel [2 ‖ 2]

$$\frac{2\times 2}{(2+2)} = \frac{4}{4} = 1\Omega$$

Step3:-

3Ω and 1Ω are in series

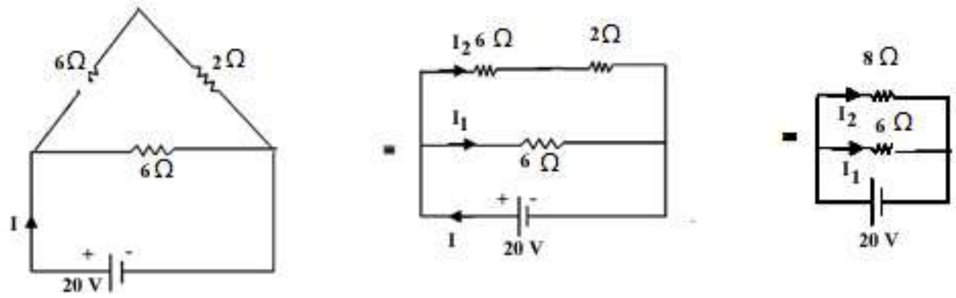
$$\therefore\ 3+1 = 4\ \Omega$$



Step4:

4Ω and 4Ω are in parallel [4 ‖ 4]

$$= \frac{4\ \times 4}{4+4} = \frac{16}{8} = 2\Omega$$



Step5:

$$I_1 = \frac{20}{8} = 3.33\text{A}$$

$$I_2 = \frac{V}{R_2} = \frac{20}{8} = 2.5\text{A}$$

$$\therefore I = I_1 + I_2$$

$$= 3.33 + 2.5$$

$$I = 5.83A$$

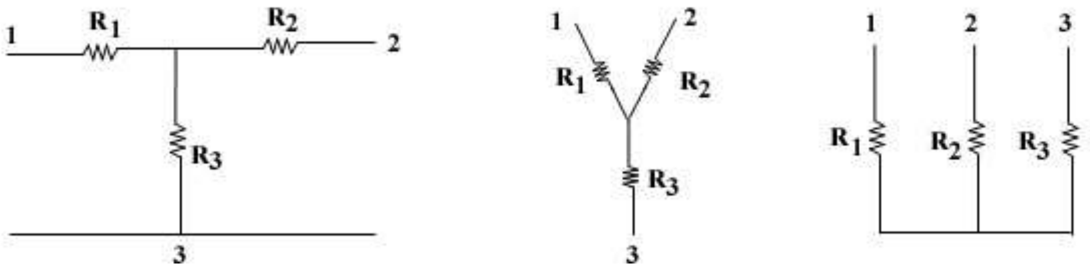
$$\text{Power dissipated in } 6\ \Omega = \text{I}_1^2 \times 6 = (3.33)^2 \times 6$$

$$= 66.53\ \text{w}$$

1.12 Star and Delta Transformation:

(i) Star Connected Circuits:

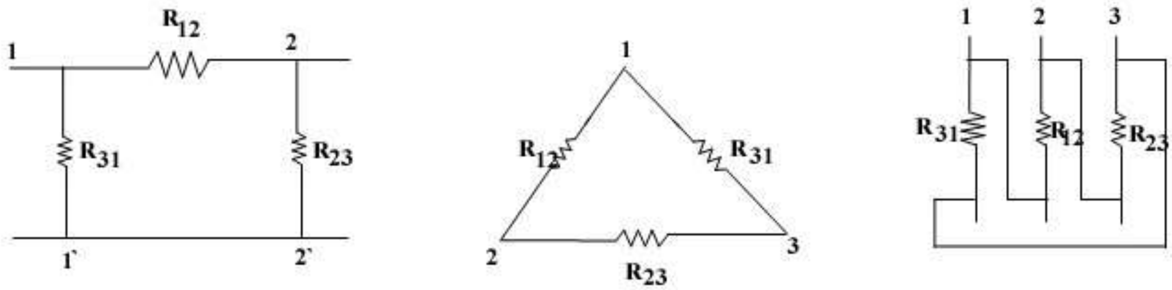
The linear lumped passive three terminal circuit consisting of three resistance R_1, R_2 and R_3 in the form of Star, Y or T connection as shown in the figure are calles as Star-Connected Circuits. They are characterised by a common node.



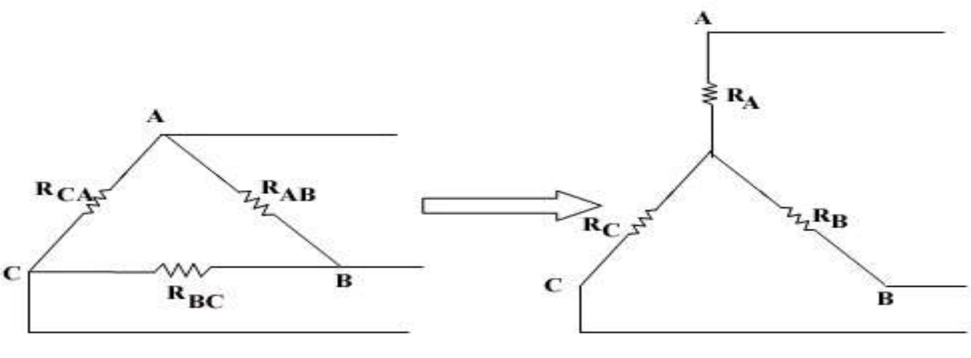
Star-Connected Circuit

(ii) Delta-Connected Circuits

The linear lumped passive three-terminal circuit consisting of three resistances R_1, R_2 and R_3 in the form of delta, Π or Δ is called as Delta-Connected Circuits.



1.13 Delta - Star Transformation:



- Referring to the Delta network,

$$\text{Resistance between A \& } B = \frac{R_{AB}\,||(R_{BC} + R_{CA})}{R_{AB} + R_{BC} + R_{CA}} \qquad (1)$$

- Referring to the Star network,

$$\text{Resistance between A \& } B = R_A + R_B \qquad (2)$$

Since the two networks are electrically equivalent, equate 1 and 2,

$$R_A + R_B = \frac{R_{AB} \,||(R_{BC} + R_{CA})}{R_{AB} + R_{BC} + R_{CA}} \qquad (3)$$

Similarly at terminals BC and CA,

$$R_B + R_C = \frac{R_{BC} \,||(R_{AB} + R_{CA})}{R_{AB} + R_{BC} + R_{CA}} \qquad (4)$$

$$R_C + R_A = \frac{R_{CA} \,||(R_{AB} + R_{BC})}{R_{AB} + R_{BC} + R_{CA}} \qquad (5)$$

Subtract equation 3 and 4,

$$(R_A + R_B) - (R_B + R_C) = \frac{R_{AB} \,||(R_{BC} + R_{CA})}{R_{AB} + R_{BC} + R_{CA}} - \frac{R_{BC} \,||(R_{AB} + R_{CA})}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_A + \cancel{R_B} - \cancel{R_B} - R_C = \frac{\cancel{R_{AB}\ R_{BC}} + R_{AB}\ R_{CA} - \cancel{R_{BC}\ R_{AB}} - R_{BC}R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_A - R_C = \frac{R_{AB}\ R_{CA} - R_{BC}R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_A - R_C = \frac{R_{CA}(R_{AB} - R_{BC})}{R_{AB} + R_{BC} + R_{CA}} \qquad (6)$$

Adding equation 5 and 6,

$$\cancel{R_C} + R_A + R_A - \cancel{R_C} = \frac{R_{CA} \,||(R_{AB} + R_{BC})}{R_{AB} + R_{BC} + R_{CA}} + \frac{R_{CA}(R_{AB} - R_{BC})}{R_{AB} + R_{BC} + R_{CA}}$$

$$2R_A = \frac{R_{CA}\ R_{AB} + \cancel{R_{CA}R_{BC}} + R_{CA}R_{AB} - \cancel{R_{CA}R_{BC}}}{R_{AB} + R_{BC} + R_{CA}}$$

$$\cancel{2}R_A = \frac{\cancel{2}R_{CA}\ R_{AB}}{R_{AB} + R_{BC} + R_{CA}}$$

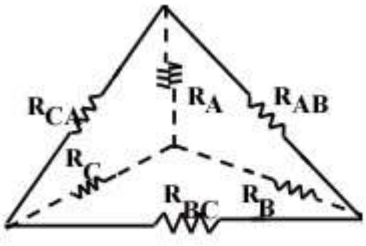
$$R_A = \frac{R_{CA}\ R_{AB}}{R_{AB} + R_{BC} + R_{CA}} \qquad (7)$$

Similarly,

$$R_B = \frac{R_{AB}\ R_{BC}}{R_{AB} + R_{BC} + R_{CA}} \qquad (8)$$

$$R_C = \frac{R_{AC}\ R_{BC}}{R_{AB} + R_{BC} + R_{CA}} \qquad (9)$$

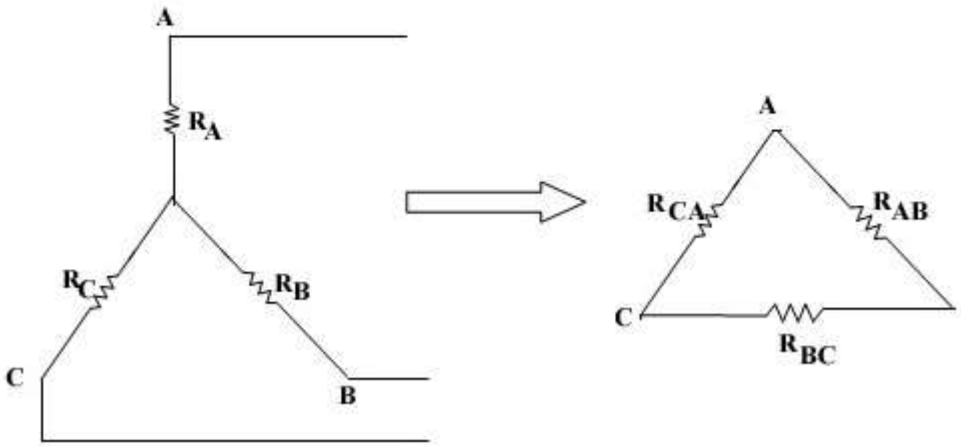
Therefore,



$$R_A = \frac{R_{AB}\ R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$
$$R_B = \frac{R_{AB}\ R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$
$$R_C = \frac{R_{AC}\ R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$

$$\text{Any arm of Star connection} = \frac{\text{Prod. of two adjacent arms of Delta network}}{\text{Sum of arms of Delta network}}$$

1.14 Star – Delta Transformation [April - 2009]



We know that,

$$R_A = \frac{R_{AB} + R_{CA}}{R_{AB} + R_{BC} + R_{CA}} \qquad (1)$$

$$R_B = \frac{R_{BC} + R_{AB}}{R_{AB} + R_{BC} + R_{CA}} \qquad (2)$$

$$R_C = \frac{R_{CA}\ R_{BC}}{R_{AB} + R_{BC} + R_{CA}} \qquad (3)$$

Multiply equation 1 and 2,

$$R_A\,.\,R_B = \frac{R_{AB}.\,R_{CA}}{R_{AB} + R_{BC} + R_{CA}} * \frac{R_{BC}.\,R_{AB}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_A\,.\,R_B = \frac{{R_{AB}}^2 R_{BC}\ R_{CA}}{(R_{AB} + R_{BC} + R_{CA})^2} \qquad (4)$$

Multiply equation 2 and 3,

$$R_B . R_C = \frac{R_{BC} + R_{AB}}{R_{AB} + R_{BC} + R_{CA}} * \frac{R_{CA}\, R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$

$$\therefore R_B \, . R_C = \frac{{R_{BC}}^2 R_{AB}\, R_{AC}}{(R_{AB} + R_{BC} + R_{CA})^2} \quad (5)$$

Multiply equation 1 and 3,

$$R_A . R_C = \frac{R_{AB} + R_{AC}}{R_{AB} + R_{BC} + R_{CA}} * \frac{R_{AC}\, R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$

$$\therefore R_A \, . R_C = \frac{{R_{AC}}^2 R_{AB}\, R_{BC}}{(R_{AB} + R_{BC} + R_{CA})^2} \quad (6)$$

Add equation 4, 5 and 6,

$$R_A \, . R_B + R_B \, . R_C + R_A . R_C = \frac{{R_{AB}}^2 R_{BC}\, R_{CA} + {R_{BC}}^2 R_{AB}\, R_{AC} + {R_{AC}}^2 R_{AB}\, R_{BC}}{(R_{AB} + R_{BC} + R_{CA})^2}$$

$$= \frac{R_{AB} R_{BC} R_{CA} (\cancel{R_{AB} + R_{BC} + R_{CA}})}{(R_{AB} + R_{BC} + R_{CA})^{\cancel{2}}}$$

$$= \frac{R_{AB} R_{BC} R_{CA}}{(R_{AB} + R_{BC} + R_{CA})} \qquad (7)$$

We know that,

$$R_B = \frac{R_{AB} + R_{BC}}{R_{AB} + R_{BC} + R_{CA}} \qquad (8)$$

Sub. Equation 8 in 7,

$$R_A R_B + R_B R_C + R_C R_A = R_B \, . R_{CA}$$

$$\therefore R_{CA} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_B}$$

$$R_{CA} = \frac{\cancel{R_B}[R_A + R_C]}{\cancel{R_B}} + \frac{R_C R_A}{R_B}$$

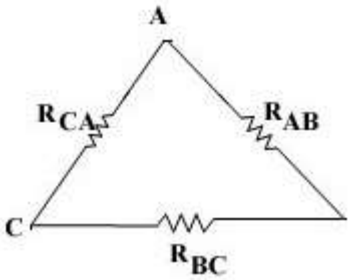
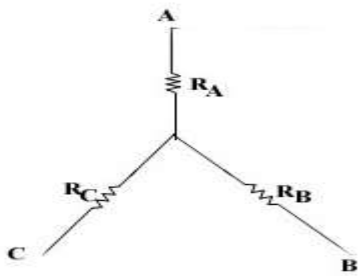
$$\therefore R_{CA} = R_A + R_C + \frac{R_C R_A}{R_B}$$

Similarly,

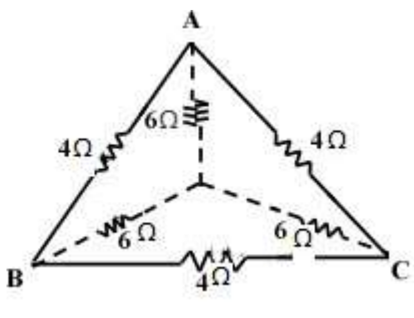
$$R_{AB} = R_A + R_B + \frac{R_A R_{AB}}{R_C}$$

$$R_{BC} = R_B + R_{CB} + \frac{R_B R_C}{R_A}$$

1.15 Difference between Star and Delta Connection

Delta Connection	Star Connection
(i) If the Delta network, effective conductance between A and B is $G_{AB} = G_{AB} + G_{CA}$	In Star network conductance is $G_{AB} = \frac{G_{AB}(G_B + G_C)}{G_A + G_B + G_C}$
(ii) it is also known as Δ - network	it is also known as Y - network
(iii) in delta connection, three resistances are connected end to end.	in star connection, one end of each resistance is connected at a point called Star-point.
(iv) it is of delta shape, so it is known as delta connection.	it appears as a star so it is known as Star connection.
(v) it is represented as 	it is represented as 

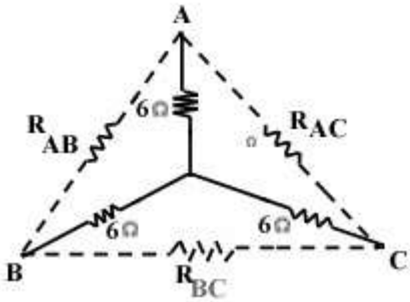
1.32 Calculate the equal resistance across AB of the circuit by using Star and Delta transformation [Nov. 2011]



Step1

The above circuit is a combination of Star and Delta network. Always star reducing the circuit from the inner side.

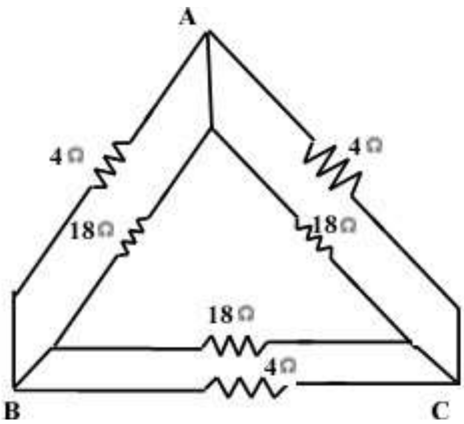
- Take the star network and convert it into Δ - network.



$$R_A = 6\ \Omega$$
$$R_B = 6\ \Omega$$
$$R_C = 6\ \Omega$$
$$R_{AB} = R_A + R_B + \frac{(R_A R_B)}{R_C}$$
$$= 6 + 6 + \frac{(6 * 6)}{6}$$
$$= 6 + 6 + 6$$
$$R_{AB} = 18\ \Omega$$
$$R_{BC} = R_B + R_C + \frac{(R_B R_C)}{R_A}$$
$$R_{BC} = 18\ \Omega$$
$$R_{CA} = R_C + R_A + \frac{(R_C R_A)}{R_B}$$
$$R_{BC} = 18\ \Omega$$

∴ The star network is converted to delta network.

∴ The resultant circuit is



Step2

In the above circuit, the resistance connected across AB i.e. 4Ω and 18Ω are in parallel connection.

Similarly the resistance across BC and CA are also parallel

∴ The above circuit can be reduced as follow,

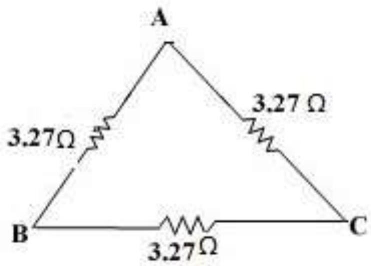
$$R_{AB} = 4\,||18$$

$$R_{AB} = \frac{4*18}{4+18} = 3.27\Omega$$

$$R_{BC} = \frac{4*18}{4+18} = 3.27\Omega$$

$$R_{CA} = \frac{4*18}{4+18} = 3.27\Omega$$

∴ The above circuit is reduced and the resultant circuit is,



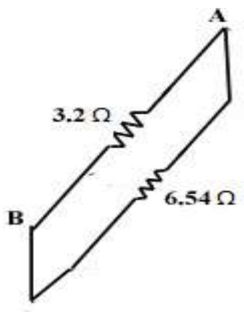
Step3:

The resistance connected across the AC and BC are in series,

$$\therefore\ R_{AC\,\&\,BC} = 3.27 + 3.27$$

$$R_{AC\,\&\,BC} = 6.54\ \Omega$$

∴The resultant circuit,



Step4:

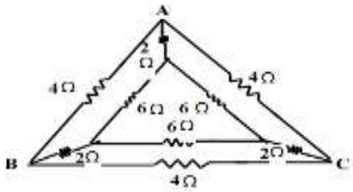
The resistance across A and B are parallel

$$\therefore\ R_{AB} = \frac{3.27*6.54}{3.27+6.54}$$

$$\therefore\ R_{AB} = 2.18\ \Omega$$

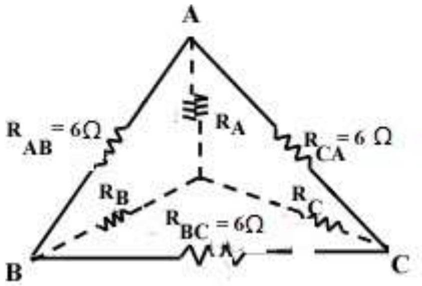
$$\therefore \text{The effective Resistance} = 2.18\ \Omega$$

1.33 Calculate the equivalent resistance across AB



Step1

Reduce the 6 Ω Δ-connected network into a star network.



$$R_A = \frac{R_{AB}.R_{CA}}{R_{AB}+R_{BC}+R_{CA}}$$

$$= \frac{6*6}{6+6+6}$$

$$\therefore\ R_A = 2\Omega$$

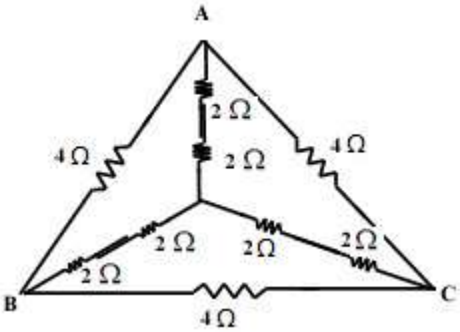
$$R_B = \frac{R_{BC}.R_{AB}}{R_{AB}+R_{BC}+R_{CA}}$$

$$\therefore\ R_B = 2\Omega$$

$$R_C = \frac{R_{CA}.R_{BC}}{R_{AB}+R_{BC}+R_{CA}}$$

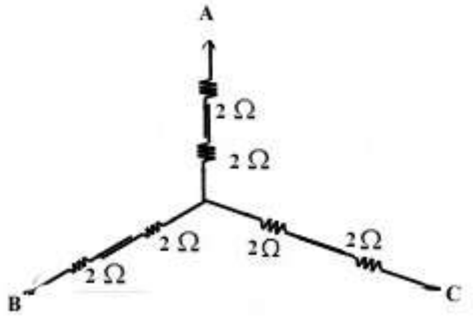
$$\therefore\ R_C = 2\Omega$$

∴ The resultant network,



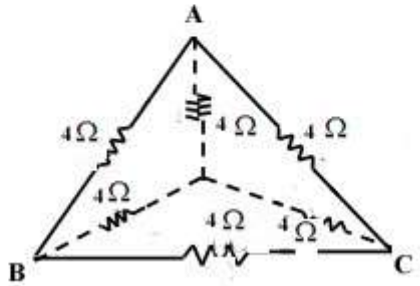
Step2

In the above circuit obtained from step 1.
Reduce the inner star connected network.
$\Rightarrow$ 2Ω and 2Ω are in series.



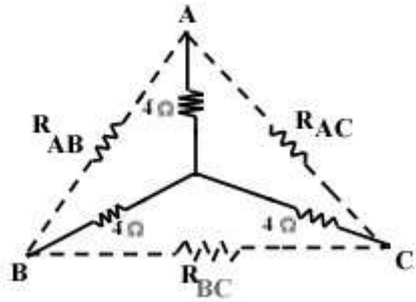
$$R_A = 2 + 2 = 4\Omega$$
$$R_B = 2 + 2 = 4\Omega$$
$$R_C = 2 + 2 = 4\Omega$$

$\Rightarrow \therefore$ The resultant circuit of step 2,



Step3

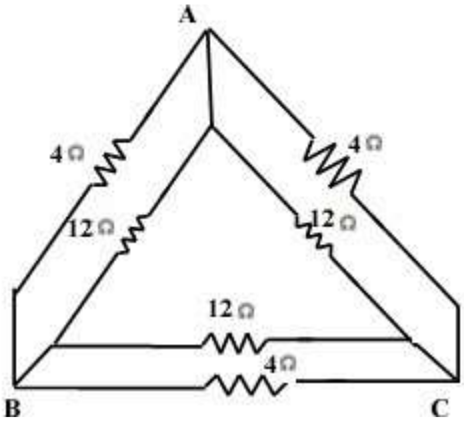
Reduce the inner star network to delta network,



$$R_{AB} = R_A + R_B + \frac{R_A \,.\, R_B}{R_C}$$
$$R_{AB} = 4 + 4 + \frac{4 * 4}{4}$$
$$\therefore R_{AB} = 12\ \Omega$$
$$R_{BC} = R_B + R_C + \frac{R_B \,.\, R_C}{R_A}$$
$$R_{BC} = 4 + 4 + \frac{4 * 4}{4}$$
$$\therefore R_{BC} = 12\ \Omega$$

$$R_{CA} = R_C + R_A + \frac{R_C \, . R_A}{R_B}$$
$$R_{CA} = 4 + 4 + \frac{4 * 4}{4}$$
$$\therefore R_{CA} = 12\ \Omega$$

$\therefore$ The resultant circuit of step 3,

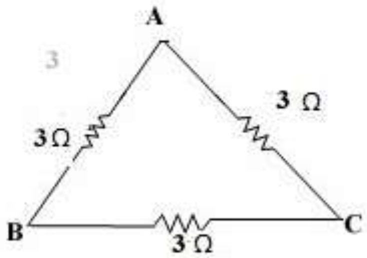


Step4

The 4 Ω and 12 Ω are in parallel in all the AB, BC, CA arms.

$$\therefore\ R_{AB} = 4\ ||12 = \frac{4 * 12}{4 + 12} = 3\Omega$$
$$R_{BC} = 4\ ||12 = \frac{4 * 12}{4 + 12} = 3\Omega$$
$$R_{CA} = 4\ ||12 = \frac{4 * 12}{4 + 12} = 3\Omega$$

$\therefore$ The resultant circuit,



Step5

Therefore the effective resistance across AB,

$$\text{Reff of } AB = \frac{R_{AB}(R_{BC} + R_{CA})}{R_{AB} + R_{BC} + R_{CA}}$$
$$= \frac{3(3 + 3)}{3 + 3 + 3}$$
$$\text{Reff of } AB = 2\ \Omega$$

1.16 *Source Transformation:*

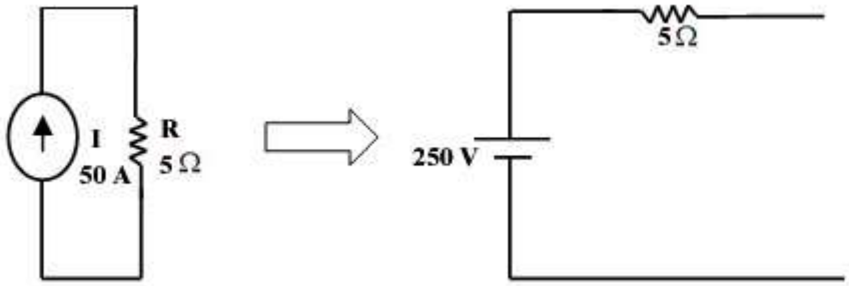
- For mesh analysis, the source should be a voltage source.

❖ For nodal analysis, the source should be a current source.

i. **Current to Voltage Transformation:**

The current source present in the mesh analysis circuit can be transformed into a voltage source by using current to voltage transformation.

i.e., consider the circuit.



$$V = IR$$
$$= 50 * 5$$
$$V = 250V$$

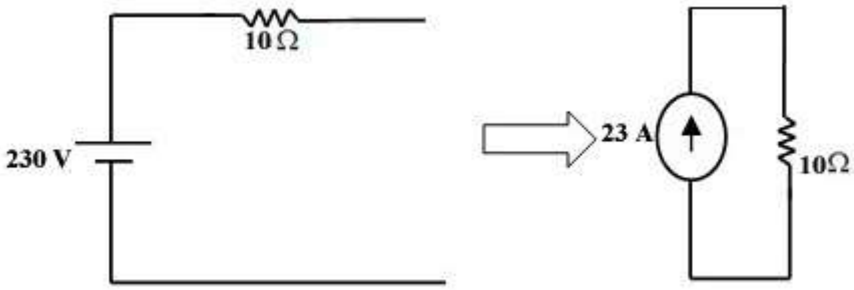
Always a current source is accompanied by resistance R which is always in parallel with the current source.

By using ohm's law, the voltage is found by using the formula $V = IR$.

While transforming a current source to voltage source, the resistance R which is parallel with the current source will now become series with the voltage source.

i.e. A voltage source is always accompanied with a series resistance.

ii. **Voltage to current source transformation**



$$I = \frac{V}{R}$$
$$I = \frac{230}{10}$$
$$I = 23 \text{ Amp}$$

1.17 Mesh and Nodal Analysis for DC circuit

1.17.1 *Maxwell's Mesh Analysis (Current Method)*

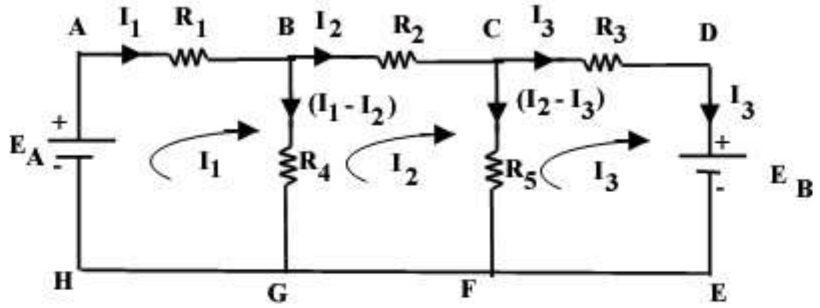
In this method, mesh or loop currents are taken instead of branch currents.

The following steps are taken while solving a network by this method:-

1. The whole network is divided into number of mesh. Each mesh is assigned a current having a continuous path. (Current is not splitted at the junction). These mesh currents are drawn in the clockwise direction. The common branch carries the algebraic sum of the mesh current flowing through it.
2. Write Kirchoff's voltage law equation for mesh using the same signs as applied to Kirchoff's law.
3. Number of equations must be equal to the number of unknown quantities. Solve the equations and determine the mesh currents.
4. From the mesh current, determine the branch currents.

Generalisation of Mesh Analysis:

Consider the network,



No. of Mesh = 3(ABGHA, BCFGB, CDEFC)

Step1:

Apply KVL loop 1, [leading current is I_1]

$$E_A - I_1R_1 - (I_1 - I_2)R_4 = 0$$
$$E_A - I_1R_1 - I_1R_4 + I_2R_4 = 0$$
$$-I_1R_1 - I_1R_4 + I_2R_4 = -E_A$$
$$-I_1(R_1 + R_4) + I_2R_4 = -E_A$$
$$I_1(R_1 + R_4) - I_2R_4 = E_A \qquad (1)$$

Apply KVL for loop 2, [leading current is I_2]

$$+(I_1 - I_2)R_4 - I_2R_2 - (I_2 - I_3)R_5 = 0$$
$$I_1R_4 - I_2R_4 - I_2R_2 - I_2R_5 + I_3R_5 = 0$$
$$I_1R_4 - (R_4 + R_2 + R_5)I_2 + I_3R_5 = 0$$
$$\therefore -I_1R_4 + I_2(R_4 + R_2 + R_5) - I_3R_5 = 0 \qquad (2)$$

Apply KVL for loop 3, [leading current is I_3]

$$+R_5(I_2 - I_3) - I_3R_3 - E_B = 0$$
$$I_2R_5 - I_3R_5 - I_3R_3 - E_B = 0$$
$$I_2R_5 - I_3(R_3 + R_5) = E_B$$
$$-I_2R_5 + I_3(R_3 + R_5) = -E_B \qquad (3)$$

The equation 1, 2 and 3 can be written in matrix form,

$$\begin{bmatrix} R_1+R_4 & -R_4 & 0 \\ -R_4 & R_4+R_2+R_5 & -R_5 \\ 0 & -R_5 & R_3+R_5 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}=\begin{bmatrix} E_A \\ 0 \\ -E_B \end{bmatrix}$$

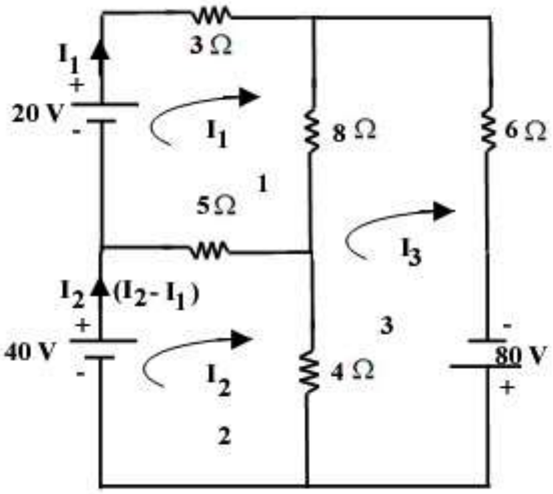
The above matrix can be written as,

$$\begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}=\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

It can be generalized as,

$$[R][I]=[V]$$

9.1 Calculate the current through the circuit using Mesh analysis? [May 2010]



No. of Mesh = 3

No. of unknown current $=3(I_1, I_2$ and $I_3)$

$\therefore$ The size of R-matrix $=3\times 3$

$\therefore$ The matrix is,

$$\begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}=\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

$$R_{11}=3+8+5=16\Omega$$

$$R_{12}=-5\Omega$$

$$R_{13}=-8\Omega$$

$$R_{21}=-5\Omega$$

$$R_{22} = 5 + 4 = 9\Omega$$
$$R_{23} = -4\Omega$$
$$R_{31} = -8\Omega$$
$$R_{32} = -4\Omega$$
$$R_{33} = 8 + 6 + 4 = 18\Omega$$

$$V_1 = 20V$$
$$V_2 = 40V$$
$$V_3 = 80V$$

$\therefore$ The matrix,

$$\begin{bmatrix} 16 & -5 & -8 \\ -5 & 9 & -4 \\ -8 & -4 & 18 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 40 \\ 80 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 16 & -5 & -8 \\ -5 & 9 & -4 \\ -8 & -4 & 18 \end{vmatrix} = 990$$

$$\Delta I_1 = \begin{vmatrix} 20 & -5 & -8 \\ 40 & 9 & -4 \\ 80 & -4 & 18 \end{vmatrix} = 15160$$

$$\Delta I_2 = \begin{vmatrix} 16 & 20 & -8 \\ -5 & 40 & -4 \\ -8 & 80 & 18 \end{vmatrix} = 19720$$

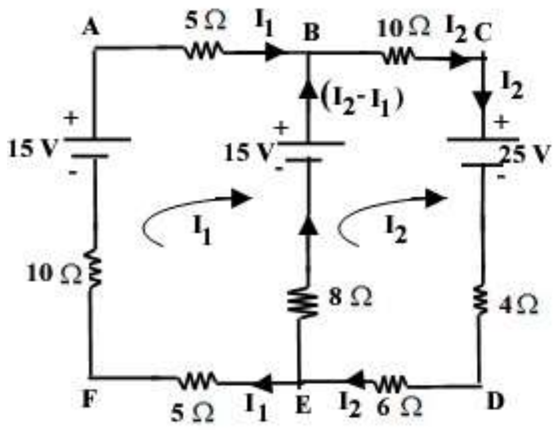
$$\Delta I_3 = \begin{vmatrix} 16 & -5 & 20 \\ -5 & 9 & 40 \\ -8 & -4 & 80 \end{vmatrix} = 15520$$

$$I_1 = \frac{\Delta I_1}{\Delta} = 15.21\text{A}$$

$$I_2 = \frac{\Delta I_2}{\Delta} = 19.91\text{A}$$

$$I_3 = \frac{\Delta I_3}{\Delta} = 15.67\text{A}$$

1.35Solve the network and find the current in each branch? [April/May 2009]

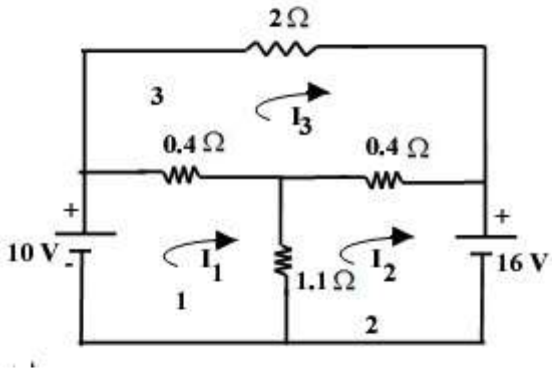


$$R_{11} = 5 + 8 + 10 = 28\Omega$$
$$R_{12} = -8\Omega$$
$$R_{22} = 10 + 4 + 6 + 8 = 28\Omega$$
$$R_{21} = -8\Omega$$
$$\begin{bmatrix} 28 & -8 \\ -8 & 28 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 40 \end{bmatrix}\begin{bmatrix} (15-15) \\ (25+15) \end{bmatrix}$$
$$= \begin{bmatrix} 28 & -8 \\ -8 & 28 \end{bmatrix} = 720$$
$$\Delta I_1 = \begin{bmatrix} 0 & -8 \\ 40 & 28 \end{bmatrix} = 320$$
$$\Delta I_2 = \begin{bmatrix} 28 & 0 \\ -8 & 40 \end{bmatrix} = 1120$$
$$I_1 = \frac{\Delta I_1}{\Delta} = 0.444A$$
$$I_2 = \frac{\Delta I_2}{\Delta} = 1.555A$$

The current in each branch,

$$AB = I_1 = 0.444A$$
$$BC = I_2 = 1.555A$$
$$AF = I_1 = 0.444A$$
$$CD = I_2 = 1.555A$$
$$ED = I_2 = 1.555A$$
$$EB = I_2 - I_1 = 1.555 - 0.444 = 1.111A$$

1.36Calculate the current in each mesh using mesh analysis [April 2008]



No. of Mesh = 3
No. of unknown current = 3(I_1, I_2, I_3)
Size of Matrix = 3 × 3

$$\begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

$R_{11} = 0.4 + 1.1 = 1.5\Omega$

$R_{12} = -1.1\Omega$

$R_{13} = -0.4\Omega$

$R_{21} = -1.1\Omega$

$R_{22} = 1.1 + 0.4 = 1.5\Omega$

$R_{23} = -0.4\Omega$

$R_{31} = -0.4\Omega$

$R_{32} = -0.4\Omega$

$R_{33} = 2 + 0.4 + 0.4 = 2.8\Omega$

$V_1 = 10V$

$V_2 = 16V$

$V_3 = 0$

$$\therefore \begin{bmatrix} 1.5 & -1.1 & -0.4 \\ -1.1 & 1.5 & -0.4 \\ -0.4 & -0.4 & 2.8 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 10 \\ -16 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 1.5 & -1.1 & -0.4 \\ -1.1 & 1.5 & -0.4 \\ -0.4 & -0.4 & 2.8 \end{vmatrix} = 2.08$$

$$\Delta I_1 = \begin{vmatrix} 10 & -1.1 & -0.4 \\ -16 & 1.5 & -0.4 \\ 0 & -0.4 & 2.8 \end{vmatrix} = 92.24$$

$$\Delta I_2 = \begin{vmatrix} 1.5 & 10 & -0.4 \\ -1.1 & -16 & -0.4 \\ -0.4 & 0 & 2.8 \end{vmatrix} = 97.04$$

$$\Delta I_2 = \begin{vmatrix} 1.5 & -1.1 & -10 \\ -1.1 & 1.5 & -16 \\ -0.4 & 0.4 & 0 \end{vmatrix} = 27.04$$

$$\therefore I_1 = \frac{\Delta I_1}{\Delta} = 44.34A$$

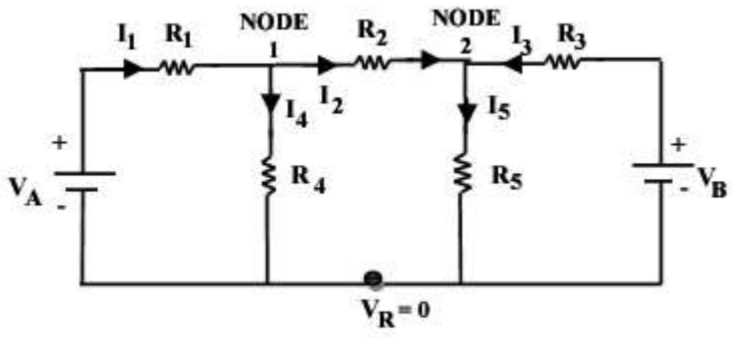
$$I_2 = \frac{\Delta I_2}{\Delta} = 46.65A$$

$$I_3 = \frac{\Delta I_3}{\Delta} = 12.99A$$

1.17.2 ***Nodal Analysis*** **[April 2011]**

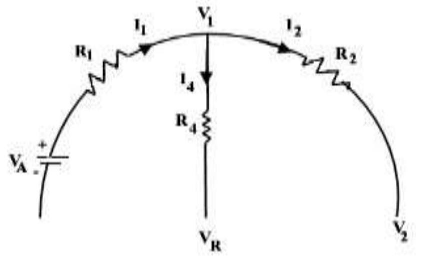
In this method, one of the nodes is taken as the reference node and the other node as independent nodes. The voltage at the different independent nodes is assumed and the equations are written as per the KCL for each node. After solving these equations, the node voltage are determined.

Consider the circuit,



- In the circuit Mark the nodes as 1 and 2.
- Apply KCL for each node.

Step1:- Apply KCL for node 1.



Sum of Incoming Current = Sum of the Outgoing Current

$$I_1 = I_2 + I_4 \qquad (1)$$

On applying Ohm's law,

$$I_1 = \frac{V_A - V_1}{R_1} \qquad (2)$$

$$I_2 = \frac{V_1 - V_2}{R_2} \qquad (3)$$

$$I_4 = \frac{V_1 - V_R}{R_4} \qquad [V_R = 0] \Rightarrow \text{ Since it is a reference node.}$$

$$I_4 = \frac{V_1}{R_4} \qquad (4)$$

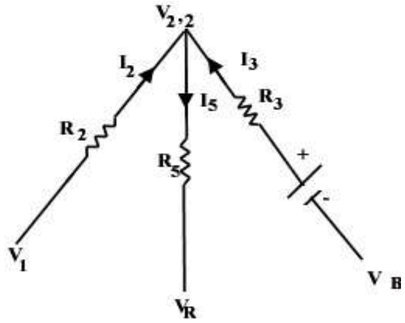
Substitute equation 2, 3 and 4 in equation 1,

$$\therefore \frac{V_A - V_1}{R_1} = \frac{V_1 - V_2}{R_2} + \frac{V_1}{R_4}$$

$$\frac{V_A}{R_1} - \frac{V_1}{R_1} = \frac{V_1}{R_2} - \frac{V_2}{R_2} + \frac{V_1}{R_4}$$

$$\frac{V_A}{R_1} = V_1\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4}\right] - V_2\left[\frac{1}{R_2}\right] \qquad (5)$$

Apply KCL at Node 2



Sum of Incoming Current = Sum of the Outgoing Current

$$I_2 + I_3 = I_5 \qquad (6)$$

On applying Ohm's law,

$$I_2 = \frac{V_1 - V_2}{R_2} \qquad (7)$$

$$I_3 = \frac{V_B - V_2}{R_3} \qquad (8)$$

$$I_5 = \frac{V_2 - V_R}{R_5} = \frac{V_2}{R_5} \qquad (9)$$

Substitute equation 7, 8 and 9 in equation 6,

$$\frac{V_1 - V_2}{R_2} + \frac{V_B - V_2}{R_3} = \frac{V_2}{R_5}$$

$$\frac{V_1}{R_2} - \frac{V_2}{R_2} + \frac{V_B}{R_3} - \frac{V_2}{R_3} = \frac{V_2}{R_5}$$

$$\frac{V_B}{R_3} = \frac{V_2}{R_5} + \frac{V_2}{R_3} + \frac{V_2}{R_2} - \frac{V_1}{R_2}$$

$$\frac{V_B}{R_3} = V_2\left[\frac{1}{R_3} + \frac{1}{R_5} + \frac{1}{R_2}\right] - V_1\left[\frac{1}{R_2}\right] \qquad (10)$$

By Matrix Method,

$$\begin{bmatrix} \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4} & -\frac{1}{R_2} \\ -\frac{1}{R_2} & \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_5} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \begin{bmatrix} \frac{V_1}{R_1} \\ \frac{V_B}{R_3} \end{bmatrix}$$

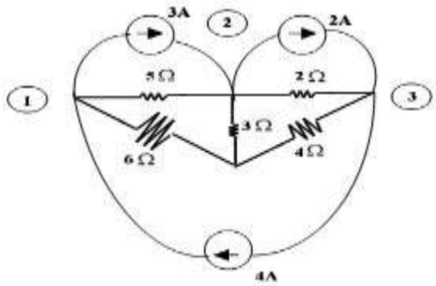
$$\frac{1}{R} = G \qquad \text{(conductance)}$$

$$\begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

In general,

$$[G][V] = [I]$$

1.37 Calculate the node voltage for the given circuit?



$G_{11} = \frac{1}{6} + \frac{1}{5} = 0.36\mho$

$G_{12} = -\frac{1}{5} = -0.2\mho$

$G_{13} = 0$

$G_{21} = -\frac{1}{5} = -0.2\mho$

$G_{22} = \frac{1}{5} + \frac{1}{2} + \frac{1}{3} = 1.03\mho$

$G_{23} = -\frac{1}{2} = -0.5\mho$

$G_{31} = 0$

$G_{32} = -\frac{1}{2} = -0.5\mho$

$G_{33} = \frac{1}{2} + \frac{1}{4} + \frac{1}{3} = 0.75\mho$

Sum of Incoming Current = Sum of the Outgoing Current

$\therefore I_1 = 4 - 3 = 1A$

$I_2 = 3 - 2 = 1A$

$I_3 = 2 - 4 = -2A$

By matrix,

$$\begin{bmatrix} 0.36 & -0.2 & 0 \\ -0.2 & 1.03 & -0.5 \\ 0 & -0.5 & 0.75 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 0.36 & -0.2 & 0 \\ -0.2 & 1.03 & -0.5 \\ 0 & -0.5 & 0.75 \end{vmatrix} = 0.1581$$

$$\Delta V_1 = \begin{vmatrix} 1 & -0.2 & 0 \\ -0.2 & 1.05 & -0.5 \\ -2 & -0.5 & 0.75 \end{vmatrix} = 0.47$$

$$\Delta V_2 = \begin{vmatrix} 0.36 & 1 & 0 \\ -0.2 & 1 & -0.5 \\ 0 & -2 & 0.75 \end{vmatrix} = 0.06$$

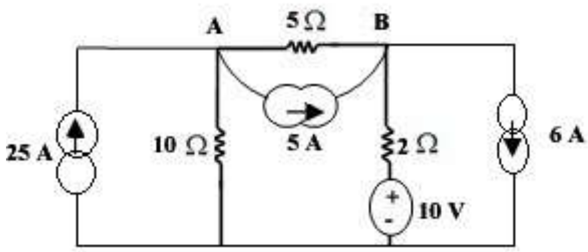
$$\Delta V_3 = \begin{vmatrix} 0.36 & -0.2 & 1 \\ -0.2 & 1.03 & -1 \\ 0 & -0.5 & -2 \end{vmatrix} = -0.3816$$

$$\therefore V_1 = \frac{\Delta V_1}{\Delta} = 2.98V$$

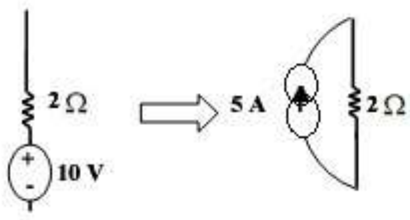
$$V_2 = \frac{\Delta V_2}{\Delta} = 0.37V$$

$$V_3 = \frac{\Delta V_3}{\Delta} = -2.41V$$

1.38 Calculate the voltage at the node A and B in the given circuit? [Jan 2010]

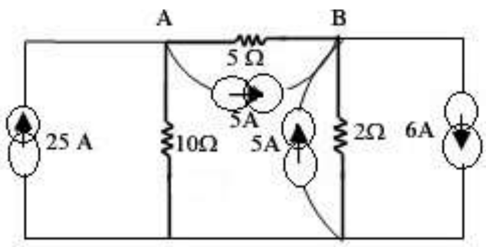


- Mark the nodes as A and B
- The voltage source in the circuit must be transformed into a current source.

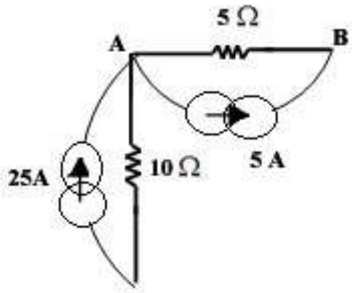


$$I = \frac{V}{R} = \frac{10}{2} = 5A$$

$\therefore$ The resultant circuit,



Apply KCL at node A,

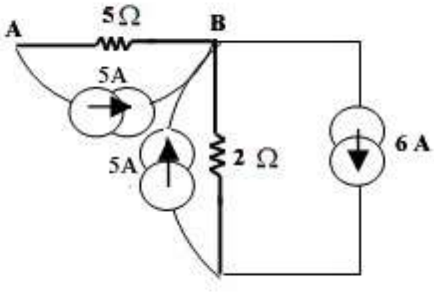


$$I_1 = 25 - 5 = 20A$$

$$G_{11} = \frac{1}{10} + \frac{1}{5} = 0.3℧$$

$$G_{12} = -\frac{1}{5} = -0.2℧$$

Apply KCL at node B,



$I_2 = \text{Incoming} - \text{Outgoing current}$
$= (5 + 5) - 6$
$= 10 - 6$
$I_2 = 4A$

$$G_{21} = -\frac{1}{5} = -0.2\mho$$

$$G_{22} = \frac{1}{5} + \frac{1}{2} = 0.7\mho$$

$$\begin{bmatrix} 0.3 & -0.2 \\ -0.2 & 0.7 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 20 \\ 4 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 0.3 & -0.2 \\ -0.2 & 0.7 \end{vmatrix} = 0.17$$

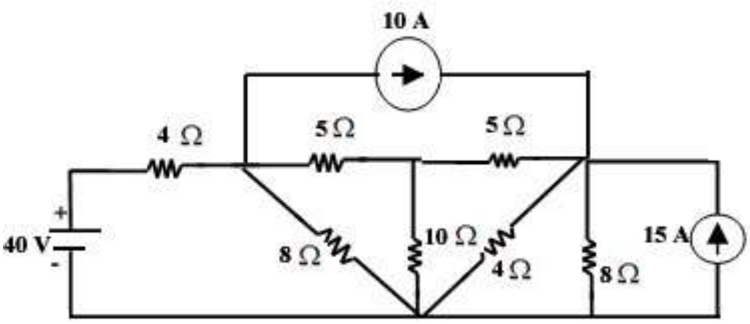
$$\Delta V_1 = \begin{vmatrix} 20 & -0.2 \\ 4 & 0.7 \end{vmatrix} = 14.8$$

$$\Delta V_2 = \begin{vmatrix} 0.3 & 20 \\ -0.2 & 4 \end{vmatrix} = 5.2$$

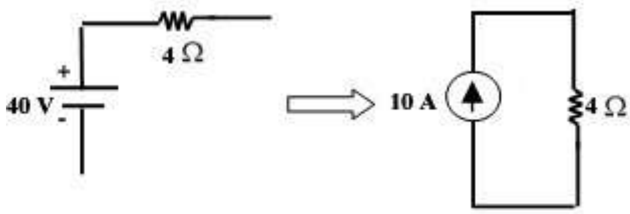
$$V_1 = \frac{\Delta V_1}{\Delta} = 87.05V$$

$$V_2 = \frac{\Delta V_2}{\Delta} = 30.5V$$

1.39 Use the Nodal Voltage method and hence find the power dissipated in 10 Ω resistor?

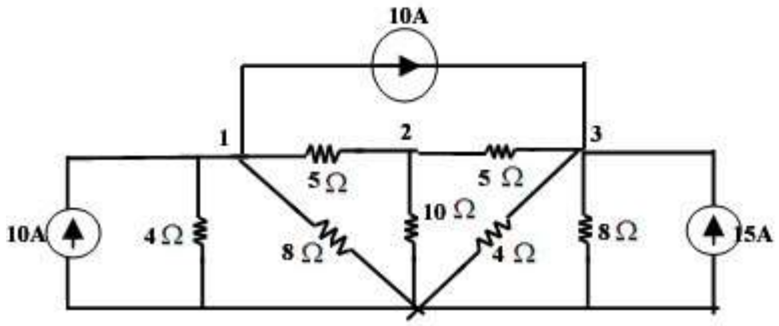


- The voltage source 40 V must be transformed into a current source.



$$I = \frac{V}{R} = \frac{40}{4} = 10\text{A}$$

$\therefore$ The resultant circuit,



Apply KCL at Node 1, 2, 3,

I_1 = Sum of Incoming Current - Outgoing Current

$$= 10 - 10$$

$$I_1 = 0A$$

$$I_2 = 0$$

$$I_3 = 15 + 10$$

$$I_3 = 25A$$

$$G_{11}=\frac{1}{4}+\frac{1}{8}+\frac{1}{5}=0.575\mho$$

$$G_{12}=-\frac{1}{5}=-0.2\mho$$

$$G_{13}=0$$

$$G_{21}=-\frac{1}{5}=-0.2\mho$$

$$G_{22}=\frac{1}{5}+\frac{1}{5}+\frac{1}{10}=0.5\mho$$

$$G_{23}=-\frac{1}{5}=-0.2\mho$$

$$G_{31}=0$$

$$G_{32}=-\frac{1}{5}=-0.2\mho$$

$$G_{33}=\frac{1}{4}+\frac{1}{5}+\frac{1}{8}=0.575\mho$$

$$\begin{bmatrix}0.575 & -0.2 & 0\\ -0.2 & 0.5 & -0.2\\ 0 & -0.2 & 0.575\end{bmatrix}\begin{bmatrix}V_1\\ V_2\\ V_3\end{bmatrix}=\begin{bmatrix}0\\ 0\\ 25\end{bmatrix}$$

$$\Delta=\begin{vmatrix}0.57 & -0.2 & 0\\ -0.2 & 0.5 & -0.2\\ 0 & -0.2 & 0.57\end{vmatrix}=0.1193$$

$$\Delta V_1=\begin{vmatrix}0 & -0.2 & 0\\ 0 & 0.5 & -0.2\\ 25 & -0.2 & 0.57\end{vmatrix}=1$$

$$\Delta V_2=\begin{vmatrix}0.57 & 0 & 0\\ -0.2 & 0 & -0.2\\ 0 & 25 & 0.57\end{vmatrix}=2.875$$

$$\Delta V_3=\begin{vmatrix}0.57 & -0.2 & 0\\ -0.2 & 0.5 & 0\\ 0 & -0.2 & 25\end{vmatrix}=6.1875$$

$$V_{10\Omega}=V_2=\frac{\Delta V_2}{\Delta}=24.09V$$

$\therefore$ Power Delivered By

$10\Omega, P_{10\Omega}=V_{10\Omega}*I_{10\Omega}$

$$\therefore P_{10\Omega}=\frac{V_{10\Omega}^2}{R_{10\Omega}}=\frac{(24.09)^2}{10}=58.03W$$